Sufficient Statistics for Economic Mobility*

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Abstract

Empirical studies of intergenerational mobility use different statistics—intergenerational elasticities, rank–rank slopes, transition matrices, axiomatic indices—and sometimes reach different conclusions about which places or periods experienced more mobility. I show that a single property of the parent–child distribution determines when all measures agree: concordance, which captures how rank-preserving the distribution is. When one economy is everywhere more rank-preserving than another, all standard relative-mobility measures produce identical orderings. Under this condition, the measures become substitutable—a researcher observing any one can infer what all others would show. I prove this holds for regression-based, transition-matrix, and axiomatic measures; remains valid under proxy outcomes and measurement error; and arises naturally in a Becker–Tomes–Loury model when parental investments weakly increase. When concordance fails to rank distributions, different mobility measures disagree because each apply different weights to regions of the joint distribution. These weights matter when concordance fails to rank distributions, typically because inequality changes across contexts. Twentieth-century U.S. data validate the predictions: rank-based measures move together while elasticity-based measures diverge as inequality rises.

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1. Introduction

Economic mobility is measured in a variety of ways—rank-rank correlations, earnings elasticities, transition matrices or axiomatic indices—which are used to document differences across cohorts, regions, demographic groups or countries.[1] Empirical work documents substantial variation in intergenerational mobility across countries (e.g., Corak 2013; Jäntti et al. 2006), regions (e.g., Chetty et al. 2014a; Corak 2020), and over historical periods (e.g., Feigenbaum 2018; Collins and Wanamaker 2022). While different measures often produce consistent rankings of mobility (e.g., Katz and Krueger 2017, Berman 2022, Deutscher and Mazumder 2023), some prominent cases—including long-run U.S. trends (Jácome, Kuziemko, and Naidu 2025) and cross-country comparisons (Blanden 2013)—yield conflicting conclusions. This raises two fundamental questions: when should we expect standard mobility measures to deliver the same ranking of economies, and when should disagreement be interpreted as substantive or indicative of underlying economic mechanisms?

This paper proposes a framework to organize these agreements and disagreements. When the parent–child joint distribution in one economy is everywhere more rank preserving—i.e., when parent and child ranks become more closely aligned throughout the distribution—all commonly used relative-mobility measures are *sufficient statistics* for overall mobility and become effectively substitutable: a researcher that observes one measure will obtain the same mobility ordering of economies as a second researcher that observes any other measure. This includes intergenerational elasticities, rank–rank slopes, transition-matrix indices in the tradition of Shorrocks (1978), and axiomatic measures. This observation helps to rationalize why studies that report multiple measures, either as distinct objects of interest or robustness, typically find broadly consistent results.

This property is not an arbitrary statistical restriction. Applied mobility research already rules out precisely the kinds of rank-reversing patterns that violate the order I study. Bounding exercises for absolute mobility impose the "*intuitive requirement that*

[1] Researchers use mobility metrics to study the dependence of income across and within generations (e.g., Solon 1992; Chetty et al. 2014b), as well as wealth (e.g., Boserup, Kopczuk, and Kreiner 2018; Fagereng et al. 2020; Fagereng, Mogstad, and Rønning 2021; Audoly et al. 2024), consumption (e.g., Jappelli and Pistaferri 2006), health (e.g., Halliday 2023; Black et al. 2024), education (e.g., Black and Devereux 2011), socio-emotional skills (Attanasio, De Paula, and Toppeta 2025), or occupational prestige (Mazumder and Acosta 2015; Olivetti and Paserman 2015; Song et al. 2020; Haeck and Laliberté 2025). They are used in positive analysis, normative analysis (e.g., Shorrocks 1978; Atkinson and Bourguignon 1982), evaluating the impacts of social policy (e.g., Chetty and Hendren 2018a,b), and as validation or targets for structural models of economic behaviour (e.g., Abbott et al. 2019; Bolt et al. 2025; Daruich 2018).

children from higher-income families are less likely to have lower incomes" (Chetty et al. 2017 pg. 401, see also Berman 2022; Manduca et al. 2024) to rule out rank-reversing patterns. Work using transition matrices routinely treats substantial anti-diagonal mass—where high-income parents systematically produce low-income children and vice versa—as empirically implausible (Shorrocks 1978; Jäntti and Jenkins 2015). I formalize this shared practice as a uniform, rank-improving shift of the copula, similar to increases in positive assortative matching (e.g., Becker 1973) and already implicit in applied mobility research. They correspond to a single, testable dependence order on the parent–child copula, known as concordance (Tchen 1980). This yields a simple insight: increasing concordance goes hand in hand with decreasing mobility by any standard measure.

This substitutability has immediate practical implications. Researchers have advanced different reasons for preferring one measure over another: elasticities are interpretable, model-consistent and widely reported (Black and Devereux 2011), but sensitive to life-cycle bias (Haider and Solon 2006); rank–rank slopes are robust and unit-free (Chetty et al. 2014a); and transition matrices are transparent (Jäntti and Jenkins 2015) with minimal data requirements, but depend on discretization (Cowell and Flachaire 2018). When economies can be ordered by the rank-strengthening copula shift above, these debates are moot: all measures deliver the same ranking.

The concordance order is both empirically testable and theoretically grounded in standard models of human capital investment. Measuring mobility with increased concordance is directly testable using standard inequality or moment-inequality tests, and it is automatically satisfied in many parametric joint distributions commonly used in applied work. Moreover, the same rank-strengthening pattern is generated by standard intergenerational investment models. In a Becker and Tomes (1979)-Loury (1981) framework, economic primitives that raise the return to human capital or improve the skill-formation technology shift optimal investment outward at every parental income level. I show that such uniform investment shifts induce exactly the copula changes described above, so all of the usual mobility measures decline together. This identifies a clear class of model-based shocks for which the empirical "*measures-as-substitutes*" result applies, regardless of whether those shocks occur through educational expansion, technological change, or policy reforms.

While this covers many canonical comparative statics, concordance does not hold universally. Not all economically relevant changes take this form including policies or shocks that improve outcomes for some parts of the distribution while weakening the parent–child link elsewhere: for instance, targeted interventions for disadvantaged families that coincide with increased sorting among elites. In such cases, different mobility measures emphasize different regions of the joint distribution, leading to disagreement. The divergence between elasticity and rank-based measures documented for the United States over the 20th century (Jácome, Kuziemko, and Naidu 2025), for example, reflects rising income inequality in the lower tail: a change in marginal distributions that affects the intergenerational elasticity but doesn't affect rank-based measures. Importantly, the concordance order's implications are robust to measurement error and the use of proxies (such as education or occupation) in place of lifetime income, strengthening the practical value of these sufficiency results. The framework in this paper makes such disagreements transparent by tracing them to the parts of the copula on which each measure loads, and clarifies when divergence reflects changing inequality rather than changing dependence structure. Thus, we also gain deeper understanding of the sources of index disagreement.

Section 7 applies the framework to twentieth-century U.S. mobility trends. The theory's predictions are borne out in the data: rank-based measures (rank-rank slopes, transition probabilities, directional mobility) move together in a wave-like pattern, while level-based measures (IGE) diverge as inequality rises. This is exactly as the framework predicts when marginal distributions change. This coherence holds despite institutional changes, varying data quality, and the use of imputed parental incomes, validating concordance as an empirically relevant organizing principle.

This paper is organised as follows. Section 2 introduces ranks, copulas, and the concordance order. Section 3 establishes that widely used mobility measures are monotone in concordance. Section 4 examines cases where concordance fails to rank two economies, explains why measures disagree, and proposes axiomatic completions of the partial order. Section 5 extends results to settings with changing marginal distributions and clarifies which measures remain robust. Section 6 microfounds concordance in a model of endogenous human capital formation. Section 8 concludes.

1.1. Related Literature

A large body of work proposes and estimates mobility measures (surveys include Black and Devereux 2011, Deutscher and Mazumder 2023). These studies ask: which measure best captures mobility? This paper asks the reverse question: under what conditions do these measures deliver the same ordering of economies or when do different measures necessarily agree? The closest work is D'Agostino and Dardanoni (2009), who, in a discrete transition-matrix setting, axiomatise concordance as a measure of rank mobility and provide a cardinalisation by scaling against a maximal benchmark. Atkinson and Bourguignon (1982) study how dependence shapes utilitarian welfare rankings in multidimensional inequality, using restrictions on the joint distribution of multidimensional economic status that are formally similar to those employed here. This paper differs in three respects. First, instead of proposing a new index, I show that concordance serves as a common dependence condition that links several mobility measures already used in applied work (regression-based, transition-matrix, and axiomatic). Second, I extend the results beyond discrete matrices to continuous joint distributions and to environments with proxies and classical measurement error. Third, I provide a microeconomic interpretation in a Becker-Tomes-Loury–type model, which makes explicit which economic forces generate concordance-increasing shifts. Whereas D'Agostino and Dardanoni focus on axiomatizing a particular mobility index and Atkinson and Bourguignon on welfare comparisons in multidimensional settings, I study the positive properties of existing empirical estimands and the conditions under which they can be treated as substitutes.

I focus on the properties and correlational structure of *estimands* of intergenerational mobility rather than their estimators. A related strand of the literature considers inference and the role of measurement error for rank-rank specifications (e.g., Chetverikov and Wilhelm 2023; Kitagawa, Nybom, and Stuhler 2018), identification under general family tree structures (e.g., Collado, Ortuño-Ortín, and Stuhler 2023; Espín-Sánchez, Ferrie, and Vickers 2023), and biases in standard estimators (e.g., Haider and Solon 2006; Nybom and Stuhler 2017). My contribution is complementary. I characterize when population parameters from different measures deliver identical orderings, regardless of how those parameters are estimated. Section 3 shows that concordance orderings are preserved under measurement error. Thus, sufficiency results apply even when researchers observe noisy proxies, although the inference challenges highlighted in this literature remain.

I focus on bivariate parent-child joint distributions with a single outcome for each generation. Researchers may observe richer data with a vector of life cycle incomes (Boserup, Kopczuk, and Kreiner 2018) , multiple generations (Adermon, Lindahl, and Palme 2021) or multiple parents as inputs (Althoff, Gray, and Reichardt 2025). Orthant orders (Shaked and Shanthikumar 2007) generalize the concordance order to multivariate settings and Audoly et al. (2024) adopt hierarchical clustering to summarize 25-dimensional outcomes. This paper's bivariate focus is deliberate. Doing so isolates the core dependence structure studied in most empirical work and permits cleaner exposition of sufficiency results.

This paper studies relative (exchange) mobility, dependence between parent and child positions, and is silent on absolute mobility measures such as the fraction of children who exceed the median income in the parent generation (Katz and Krueger 2017). Ray and Genicot (2023) show that panel independent mobility measures can be constructed when welfare depends only on growth progressivity, not origin-destination identities. The concordance framework complements this work: while Ray and Genicot characterize planner preferences over upward mobility, I characterize the dependence structure determining relative positional mobility. While some absolute mobility bounds (Chetty et al. 2017; Berman 2022; Manduca et al. 2024) impose copula restrictions, concordance clarifies when those restrictions are empirically plausible.

This paper relates to work studying sorting through concordance properties. Gola (2021), Anderson and Smith (2024) and Boerma et al. (2023) derive concordance orderings as the equilibrium outcome of assignment problems and use them, to measure sorting in the labour market. Chiappori et al. (2025) axiomatise an odds-ratio index for assortative mating that is both necessary and sufficient for concordance in discrete contingency tables and compare with a normalized-trace index that is a direct transformation of the Shorrocks (1978) mobility measure. I apply similar tools to a different question. Rather than measuring the degree of sorting between agents at a point in time, I study how dependence between generations shapes the ordering of commonly used mobility measures. Section 6 shows concordance arises endogenously in a Becker-Tomes-Loury investment model, paralleling endogenous concordance in matching models, and linking to work studying causal impacts on intergenerational mobility (e.g., Chetty et al. 2025; Nakamura, Sigurdsson, and Steinsson 2022).

2. Setting and Concordance Ordering

This paper considers the following setting. A researcher is comparing economic mobility between economies. Specifically, the focus is on the joint behavior of parent–child (or period-to-period) outcomes and how that dependence shapes measures of mobility. I begin by defining the relevant joint distributions and providing the definition of the concordance order.

Let (Y_K, Y_P) denote child and parent outcomes with joint CDF $F(Y_K, Y_P)$ and marginals $F_K(Y_K)$ and $F_P(Y_P)$. The key insight is that relative mobility—how children's positions relate to parents' positions—depends entirely on the dependence structure between these outcomes, not their levels or dispersion.

This dependence structure is captured by the copula. Define ranks as $R_K = F_K(Y_K)$ and $R_P = F_P(Y_P)$, both uniformly distributed on $[0, 1]$. By Sklar's Theorem (1959), the joint distribution can be decomposed as

$$F(Y^K, Y^P) = C(F_K(Y^K), F_P(Y^P)), \tag{1}$$

where $C : [0, 1]^2 \to [0, 1]$ is the copula—the joint distribution of ranks. The *copula* alone encodes all rank-dependence between generations, independently of marginal distributions.[2] This separation means mobility measures operating on relative positions are simply alternative summaries of the same underlying copula.[3]

To isolate dependence, the majority of the analysis will fix marginal distributions and focus solely on changes in the copula. For instance, imagine a thought experiment in which the U.S. marginal income distributions remain constant, but the parent–child rank-dependence structure varies. Under this restriction, each mobility measure becomes an alternative summary of the same copula—that is, of how often "high" outcomes for one generation coincide with "high" outcomes for the next. Section 5 shows how relaxing the constant-marginals assumption affects the results.

2.1. The Concordance Order

I begin by illustrating changes in dependence with a simple example, then formalize the concordance order.

Figure 1 presents a grid of parent–child quintile transition matrices.[4] Moving rightward or downward increases concordance: probability mass shifts from anti-diagonal (rank-reversing) to diagonal (rank-preserving) cells. For instance, the top-left panel shows perfect discordance, the center shows independence, and the bottom-right shows perfect concordance.

The statistical property of concordance formalizes the intuitive idea that "large" values of one random variable tend to coincide with "large" values of another: a form

[2] I treat the copula and marginals as observed. Alternatively, one can consistently estimate the relevant moments and I abstract from inference which can be non-trivial for rank-based measures (see Chetverikov and Wilhelm 2023). Appendix A discusses uniqueness of the copula and provides regulatory conditions for discrete cases.

[3] In addition, this property is exploited by Chetty et al. (2017), and Manduca et al. (2024) who impose copula restrictions to bound absolute mobility when panel data are unavailable. I exploit the same logic: summarizing relative mobility via the copula alone. Copulas also accommodate mass points (e.g., top-coded incomes or non-participation).

[4] These examples assume outcomes take on discrete values for simplicity.

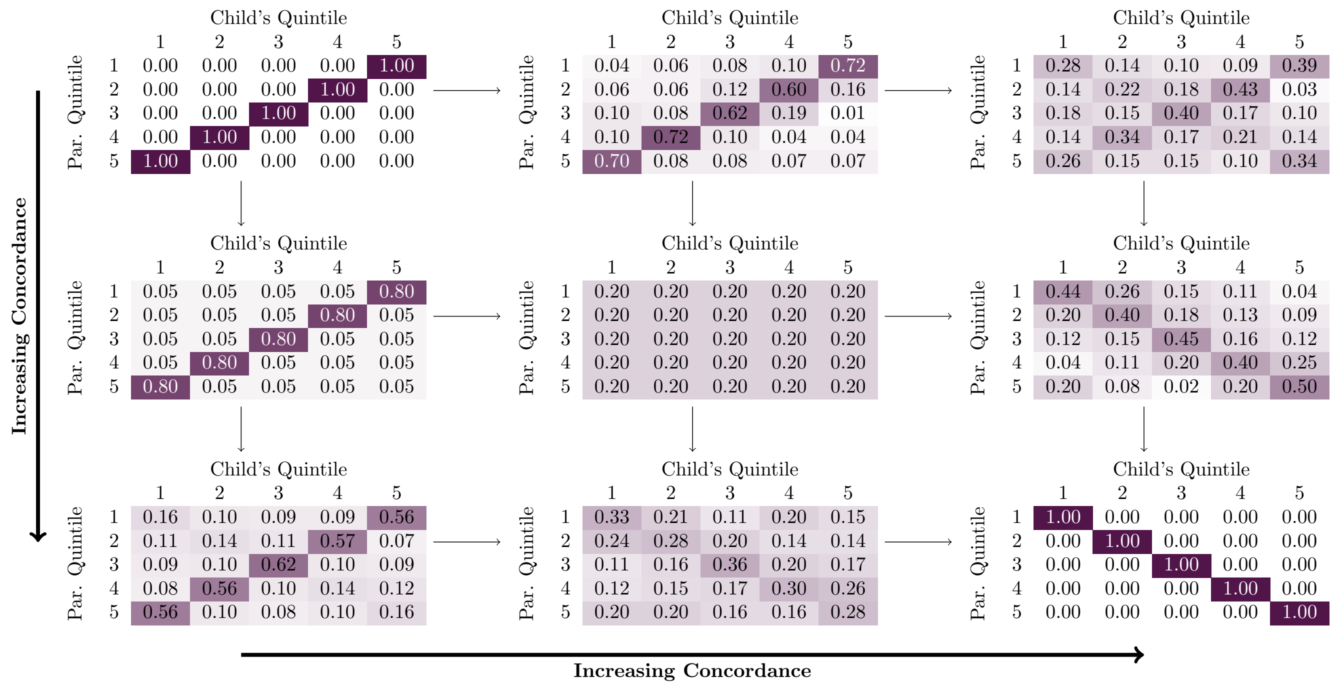


FIGURE 1. Concordance Ordering Among Transition Matrices

Notes: Moving rightward or downward concentrates probability mass from off-diagonal (rank-reversing) to diagonal (rank-preserving) cells

of stochastic dominance for dependence in joint-distributions (e.g., Kirkegaard 2017). In the context of intergenerational mobility copulas, it is higher if parental ranks are more likely to be realised with higher child ranks. Importantly, it appeals to an implicit sense of mobility. For example, the bounding exercise in Chetty et al. (2017) assumes the copula satisfies the "*intuitive requirement that children from higher-income families are less likely to have lower incomes*" (pg. 401). Concordance formalises these types of intuitions in a mathematically precise and testable stochastic ordering over copulas:[5]

DEFINITION 1 (Concordance Ordering). *Let C^A denote the copula, or joint distribution of ranks, in economy A and C^B denote the copula in economy B. The dependence between outcomes Y^K, and Y^P is said to be larger in concordance in economy A than economy B (Yanagimoto and Okamoto 1969; Tchen 1980) if and only if $C^A(u,v) \geq C^B(u,v)\ \forall\, u,v \in [0,1]$.*

This is denoted $C^A \succeq C^B$ or $A \succeq B$.

Note that higher concordance corresponds to lower mobility, as parent-child ranks

[5]Throughout I will use superscript A and B to denote primitives and measures in different economies A and B, respectively. This ordering is sometimes alternatively known as the PQD ordering or point-wise copula ordering.

become more strongly aligned: $C^A \succeq C^B$ implies less mobility in A than B. Statistical tests of concordance are provided in Cebrián, Denuit, and Scaillet (2004) and Denuit and Scaillet (2004) or can be constructed from moment-inequality testing procedures (Andrews and Shi 2013).

Formally, the concordance order captures two distinct concepts. First, the degree of monotone dependence or the tendency of random variables to cluster around the graph of any (measurable) monotone function $Y^K = f(Y^P)$ or $Y^P = g(Y^K)$. Second, the direction of monotonicity—whether the function is monotone increasing or decreasing. The concordance order runs from maximum mobility (perfect rank-reversal between generations, the Fréchet lower bound)[6] to minimum mobility (perfect rank-preservation, the Fréchet upper bound).

The following elementary exchanges define a sequence of copula transformations ordering joint-densities by their concordance (see also Tchen 1980):

Definition 2 (Concordance Increasing Exchanges). *Let C^A be a copula. Pick $0 < u_1 < u_2 < 1$ and $0 < v_1 < v_2 < 1$, and let $\varepsilon > 0$ be small enough that the update below remains a valid copula. Define*

$$C^B(u, v) \;=\; C^A(u, v) + \varepsilon\left(\mathbf{1}_{\{u \geq u_1,\, v \geq v_1\}} + \mathbf{1}_{\{u \geq u_2,\, v \geq v_2\}} - \mathbf{1}_{\{u \geq u_1,\, v \geq v_2\}} - \mathbf{1}_{\{u \geq u_2,\, v \geq v_1\}}\right).$$

We say C^B is obtained from C^A by a concordance-increasing exchange *on the rectangle $[u_1, u_2] \times [v_1, v_2]$. This operation preserves the marginals, $C^B(u, 1) = u$ and $C^B(1, v) = v$, and moves probability mass from the two off-diagonal corners of the rectangle to the two diagonal corners. The reverse update with $\varepsilon < 0$ is concordance-decreasing.*

This operation can be confined to arbitrarily small rectangles ($u_2 - u_1 < \eta$ and $v_2 - v_1 < \eta$, for $\eta > 0$), and any concordance increase can be decomposed into a sequence of such exchanges. This provides a simple building block for increased concordance: shifting probability from the corners where parents and children have opposite relative positions (one high, one low) to corners where both are high or both are low.[7]

Families of Distributions. A large class of multivariate distributions are ordered by their concordance. While the copula approach is nonparametric, many common

[6] This is akin to maximal negative rank correlation and generalising Prais (1955)'s origin independence which corresponds to the independence copula.

[7] For finite economies with N dynasties, concordance has an equivalent discrete definition for dynasities: $A \succeq B$ if swapping any two parent-child pairs satisfies $(Y_i^K - Y_j^K)(Y_i^P - Y_j^P) > 0$.

parametric families impose concordance orderings as dependence increases (Joe 2014). Bivariate normal, elliptical, and Archimedean copulas (Clayton, Gumbel, Frank) all become more concordant as their dependence parameters increase. Similarly, multivariate extensions of the Singh and Maddala (1976) size-distribution for incomes, generalizing Champernowne (1952); Fisk (1961) and Pareto distributions have a parametric concordance order. The concordance order is stable under increasing marginal transformations and under discretization to finite transition matrices. Moreover, the order is preserved under convex combinations and weak limits: mixtures of more-concordant copulas remain more concordant, and pointwise limits of concordance-ordered sequences inherit the order (Shaked and Shanthikumar 2007).

Stronger Dependence. The majority of the results in this paper follow directly from the concordance ordering. However, two of the results require an empirically plausible and stronger regularity assumption on differences in the dependence structure:[8]

ASSUMPTION 1 (Anti-Diagonal to Diagonal Exchange in Mass). *Let $\Delta C(u, v) \equiv C^A(u, v) - C^B(u, v)$ denote the difference in the copulas A and B. $\Delta C(u, v)$ is a supermodular function: for all $u_1 \leq u_2, v_1 \leq v_2 \in [0, 1]$*

$$\Delta C(u_1, v_1) + \Delta C(u_2, v_2) \geq \Delta C(u_2, v_1) + \Delta C(u_1, v_2).$$

This regularity assumption requires mass to shift toward the diagonal consistently throughout rank-space—similar ranks (low-low, high-high) co-occur more often everywhere, not just on average—ruling out offsetting movements where concordance increases in some regions but decreases in others. It implies all sub-copulas satisfy both increased concordance and Assumption 1. For many parametric families of copulas, increasing dependence implies Assumption 1 is directly satisfied. The following lemma establishes the relation to the concordance order.

LEMMA 1 (Stronger Dependence). *All copula pairs A and B that satisfy Assumption 1 also satisfy the concordance order in Definition 1.*

Using Concordance. Having established concordance as a key and intuitive dependence concept, I now turn to showing that this single ordering determines the

[8]The need to impose some restrictions on the joint density is a common problem in mobility research. As discussed above, versions of this empirical plausibility claim are invoked by Chetty et al. (2017) and Berman (2022). Shorrocks (1978) assumes away "*empirically unlikely*" transition matrices in the form of those without quasi-maximal diagonals.

rankings produced by all commonly used mobility measures. This demonstrates that concordance ensures substitutability: when it applies, researchers need not choose between competing metrics, as they all tell the same story about relative mobility.

3. Concordance Ordering As A Sufficient Statistic For Exchange Mobility

This section demonstrates that concordance orders all major classes of exchange mobility measures—from regression-based to axiomatic indices, operating on ranks, logs, or income levels. Building on Deutscher and Mazumder (2023)'s taxonomy and axiomatic contributions such as Shorrocks (1978), I focus on exchange mobility measures that depend solely on the joint-distribution. In applied work Y is most commonly income (e.g., Chetty et al. 2014b), although other measures such as wealth (e.g., Fagereng, Mogstad, and Rønning 2021) or health (e.g., Halliday 2023) are also used. In Section 6, I show that these measures also increase with investment. These coherence results are presented sequentially; formal proofs appear in Appendix B.

3.1. Regression

I begin by analysing two of the most widely used measures: the intergenerational earnings elasticity (IGE) and rank-rank regression slope. The IGE is the value of β, obtained from the following regression

$$\ln Y_i^K = \alpha + \beta \ln Y_i^P + \epsilon_i, \tag{2}$$

and the Rank-Rank regression slope, the value of ρ obtained from the following regression

$$R_i^K = \alpha + \rho R_i^P + \epsilon_i. \tag{3}$$

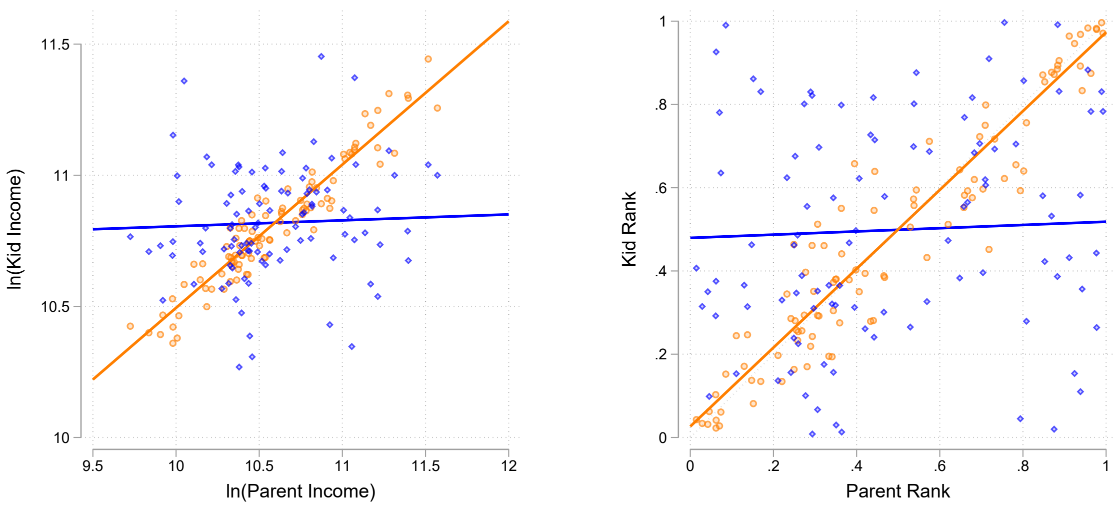
It is well known that the concordance order provides an ordering over linear regression coefficients. Thus, in the context of economic mobility, these mobility measures are unambiguously smaller in more concordant economies.

Proposition 1 (Linear Regression Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. If $C^A \succeq C^B$, then*

i. The intergenerational earnings elasticity (IGE) is ordered: $\beta^A \geq \beta^B$; and *ii. The Rank-Rank Slope is ordered:* $\rho^A \geq \rho^B$.

Both increase relative to the concordance ordering.

Figure 2 illustrates Proposition 1: as concordance increases (blue diamonds to orange circles), realizations cluster more tightly around co-monotonicity in both log- and rank-space. A direct corollary is an order over correlation measures: the intergenerational correlation, $corr(\ln Y^K, \ln Y^P)$, and the rank correlation, $corr(R^K, R^P)$ also increase relative to the concordance ordering.[9]



(A) Intergenerational Elasticity (B) Rank-Rank Regressions

FIGURE 2. Regression Measures Under Increased Concordance

Notes: Each panel shows the effect of an increase in concordance on a mobility measure. The marginal distribution of parents income and children's income are held fixed. In each panel, the correlation parameter in the elliptical copula increase and the joint-distribution of the orange circle markers is larger than the blue diamonds in the concordance order. Both panels show the same 100 draws from the joint distribution and the population value of the mobility measure. Results correspond to Proposition 1, with the log-log regression (IGE) in panel (A) and the rank-rank correlation in panel (B).

In addition, both Proposition 1 and its corollaries can be extended to consider mobility for subsets of the population. For example, children born to parents in a specific

[9] Hart (1983) and Shorrocks (1993) propose using one minus the intergenerational correlation as a measure of mobility. Similarly, increased concordance also raises *average* non-linear persistence measures (e.g., Arellano, Blundell, and Bonhomme 2017). The logic of the argument applies in one direction because not all economies can be compared by their concordance (see Section 4). However, if we restrict attention to those that can be ranked by concordance we can invert the logic. In this case (i) and (ii) must imply $C^A \succeq C^B$ and, likewise, the remaining results.

segment of the income distribution. In these cases, the assumption of a concordance order can be relaxed to apply 'locally' in a segment of the income distribution.

Local Measures. Concordance orders standard regression estimates, but also the entire conditional linear relationship between generations and, therefore, local measures of mobility. Increased concordance rotates or tilts the entire non-parametric regression curve. Thus, lowering mobility measured by the conditional expected rank $E\left[R^K \mid R^P = r\right]$.[10] I formalise this in the following proposition.

PROPOSITION 2 (Conditional Expected Rank Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. For monotone conditional expected ranks, if $C^A \succeq C^B$ the conditional expected rank measure rotates around a point $r^\star$:*

i. It is decreasing relative to the concordance ordering below $r^\star$, $E^A\left[R^K \mid R^P = r\right] \leq E^B\left[R^K \mid R^P = r\right]\ \forall\, r \leq r^\star$, and

ii. It is increasing relative to the concordance ordering above $r^\star$, $E^A\left[R^K \mid R^P = r\right] \geq E^B\left[R^K \mid R^P = r\right]\ \forall\, r \geq r^\star$.

When the conditional expected rank is approximated parametricaly using the linear projection in equation (3), then $r^\star = 0.5$ and the conditional expected rank curve rotates around the median.

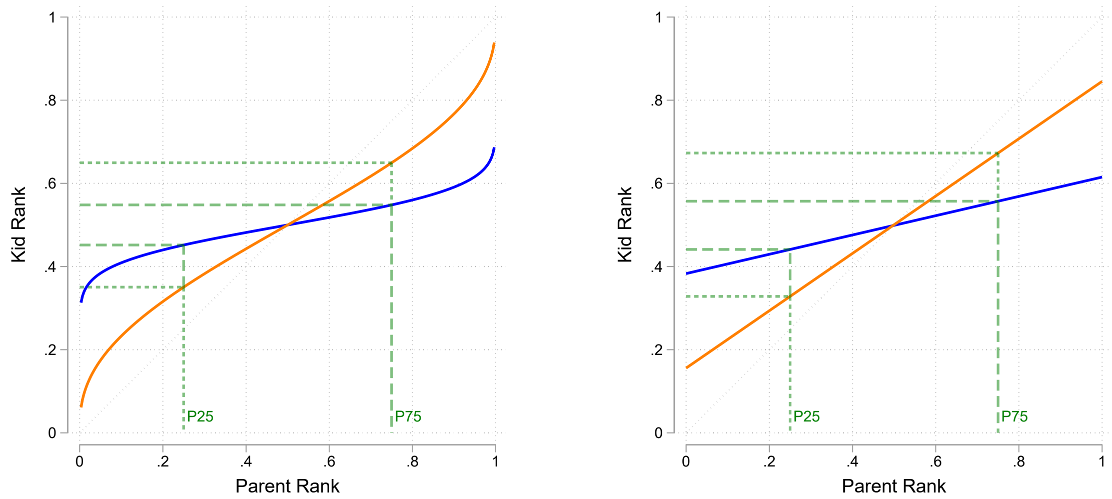
Higher concordance rotates the conditional expectation function, $E\left[R^K \mid R^P = r\right]$, toward the 45-degree line (Figure 3): children's expected rank responds more strongly to parental rank. Low-rank parents' children fall further below the median, high-rank parents' children rise further above. At perfect concordance $E\left[R^K \mid R^P = r\right] = r$ exactly as children's ranks coincide with their parents' and relative mobility vanishes.

3.2. Transition Probabilities

While regression measures summarize mobility globally, practitioners often focus on specific transitions like "rags-to-riches" or "riches-to-rags" probabilities (e.g., Corak and Heisz 1999; Chetty et al. 2014a). Concordance orders these local measures as well.

PROPOSITION 3 (Rank Based Local Measures). *Economies A and B have identical marginals, but different rank dependence denoted by copulas C^A and C^B, respectively. If $C^A \succeq C^B$, then*

[10] See Chetty et al. (2025) for a recent application using this to measure mobility.



(A) Non-Parametric

(B) Linear-Approximation

FIGURE 3. Conditional Expected Ranks Under Increased Concordance

Notes: Each panel shows the effect of an increase in concordance on a mobility measure. The marginal distribution of parents income and children's income are held fixed. In each panel, the correlation parameter in the elliptical copula increase and the joint-distribution of the orange distribution is larger than the blue distribution in the concordance order. Both panels show how the population value of the mobility measure rotates corresponding to Proposition 2. Dashed green lines correspond to the measure evaluated at the 25$^{\text{th}}$ and 75$^{\text{th}}$ percentile— commonly used in the literature.

i. Transition Probabilities: Measures along the positive diagonal are increasing relative to the concordance ordering, while those in off diagonals are decreasing:

a. Positive diagonal: $TP^A\left[R^K > \tau^k \mid R^P > \tau^p\right] \geq TP^B\left[R^K > \tau^k \mid R^P > \tau^p\right]$ *and* $TP^A\left[R^K \leq \tau^k \mid R^P \leq \tau^p\right] \geq TP^B\left[R^K \leq \tau^k \mid R^P \leq \tau^p\right]\ \forall \tau^k, \tau^p \in [0,1]$.

b. Off-diagonal: $TP^A\left[R^K > \tau^k \mid R^P \leq \tau^p\right] \leq TP^B\left[R^K > \tau^k \mid R^P \leq \tau^p\right]$ *and* $TP^A\left[R^K \leq \tau^k \mid R^P > \tau^p\right] \leq TP^B\left[R^K \leq \tau^k \mid R^P > \tau^p\right]\ \forall \tau^k, \tau^p \in [0,1]$.

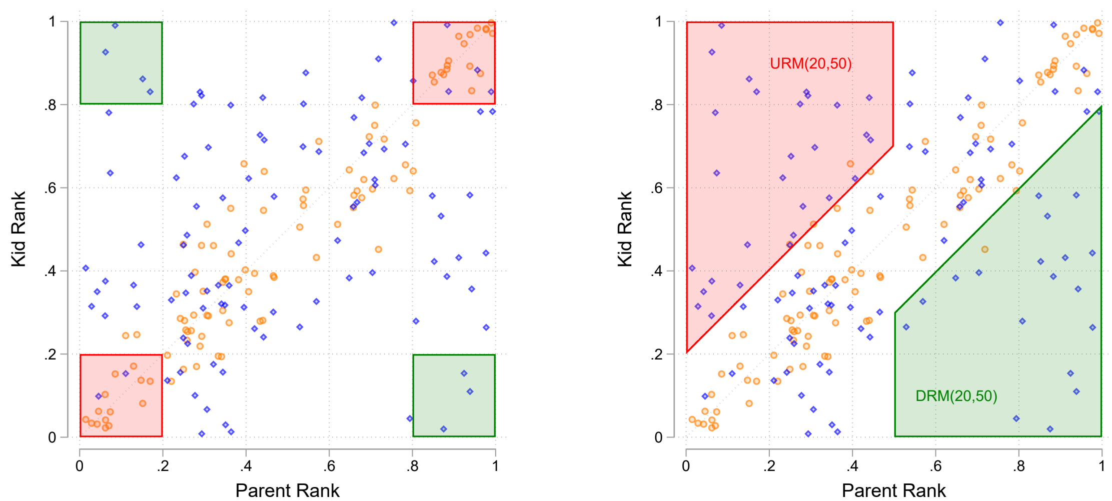
ii. Bhattacharya and Mazumder (2011)'s Directional Rank Mobility: If Assumption 1 holds then the probability a child's rank is larger than their parent's rank, by an amount s, for those with parents below τ is decreasing relative to the concordance ordering:

$$URM^B(s,\tau) \geq URM^A(s,\tau)\ \forall (s,\ \tau) \in [0,1]^2 \text{ where } URM(s,\tau) = Pr(R^K - R^P > s \mid R^P \leq \tau)$$

Analogously, Downward Rank Mobility (DRM).

Figure 4 illustrates this result: as concordance increases, the frequency of extreme transitions declines. Unlike transition probabilities which depend only on copula values at specific points, directional mobility integrates over regions and requires the stronger regularity condition that mass shifts uniformly toward the diagonal (Assumption 1).

This highlights the power of concordance as a single dependence metric: across a range of local and global mobility notions, higher concordance uniformly implies lower mobility.



(A) Transition Probabilities

(B) Directional Rank-Mobility

FIGURE 4. Rank-Space Probalities Under Increased Concordance

Notes: Each panel shows the effect of an increase in concordance on a mobility measure. The marginal distribution of parents income and children's income are held fixed. In each panel, the correlation parameter in the elliptical copula increase and the joint-distribution of the orange circle markers is larger than the blue diamonds in the concordance order. Both panels show 100 draws from the joint distribution and the population value of the mobility measure. Results correspond to Proposition 3. Panel (A) shows the off-diagonal (in green) and on-diagonal (in red) measures as regions of the rank-space using quintiles of the distribution. Panel (B) shows downwards rank mobility (in red) and upwards rank mobility as regions of the rank-space using above and below median parents who have children jumping more than a quintile in the child distribution.

3.3. Fixed Rank Comparisons

Empirical studies often evaluate ranks against a fixed external benchmark rather than within their own population. Examples include ranking children by their position in the parent cohort's income distribution; comparing subnational mobility against the national income distribution (e.g. Chetty et al. 2014a; Bütikofer, Dalla-Zuanna, and Salvanes 2022; Deutscher and Mazumder 2020; Bell, Blundell, and Machin 2023); or for comparisons of outcomes across racial groups (in the spirit of Bayer and Charles 2018).[11] Let

$$\widetilde{R}^K = F_{\widetilde{Y}}\left(F^{-1}_{Y^K}\left(R^K\right)\right), \tag{4}$$

be the child's rank in a fixed reference distribution $\widetilde{Y}$, e.g. their ranking in the parent's income distribution. Since this is strictly increasing in R^K it preserves the copula. Consequently, all rank-based exchange mobility measures remain monotone in concordance even under fixed benchmark rankings (an analogous argument applies for parent ranks). Those who with higher income ranks in their own distribution are also higher in the fixed reference distribution of incomes.

The results so far establish concordance as sufficient for comparing mobility using various empirical approaches. I now turn to axiomatic measures, which take a different methodological approach by deriving mobility indices from first principles rather than statistical relationships.

3.4. Axiomatic Measures of Exchange Mobility

A complementary approach axiomatizes mobility measures (similarly to inequality measurement). They precisely define minimal and maximal mobility as well as necessary properties of mobility measures. Axiomatic frameworks (e.g. Fields and Ok 1996; Cowell and Flachaire 2018) specify distance metrics on parent–child distributions that uniquely define a mobility index. Likewise, D'Agostino and Dardanoni (2009) derive a rank-based concordance index from ordering axioms, and Shorrocks' (1978) trace measure captures the probability of escaping one's parental "class". Under common marginals each of these indices hinges solely on the underlying dependence structure. Therefore, they all increase in the concordance order.

Proposition 4 (Axiomatic Measures). *The following axiomatic measures of exchange mobility are decreasing in the concordance order:*

i. Fields and Ok (1996, 1999)'s measures of absolute differences $\int\int|Y^K - Y^P| f\left(Y^K, Y^P\right) dY^K dY^P$ and it's decomposition into transitions.

[11] Nonparametric estimation of these fixed-benchmark ranks corresponds to absolute mobility measures (Deutscher and Mazumder 2023).

ii. *D'Agostino and Dardanoni (2009)'s concordance measure of rank mobility for global matrices and the extension .*

iii. *Cowell and Flachaire (2018)'s measure of the distance between relative positions,* $\frac{1}{\alpha(\alpha-1)} \int \int \left(\frac{Y^K}{\overline{Y}^K}\right)^{\alpha} \left(\frac{Y^P}{\overline{Y}^P}\right)^{1-\alpha} f\left(Y^K, Y^P\right) dY^K dY^P$*, where* α *controls the relative weight on upwards movements compared to downwards movements.*

Additionally, if Assumption 1 holds then

iv. *Shorrocks (1978)'s trace measure,* $\frac{q-\text{trace}(\Omega)}{q-1}$*, for stochastic matrix* Ω *with elements* $\Omega_{i,j} = Pr\left(\frac{j-1}{q} < R^K \leq \frac{j}{q} \mid \frac{i-1}{q} < R^P \leq \frac{i}{q}\right)$*.*[12]

Figure 5 shows how an increase in concordance affects these mobility measures. The distributions of absolute differences is more compressed (Panel A) and relative statuses (Panel B) fall on average, lowering Fields and Ok (1996) and Cowell and Flachaire (2018) measures. At the same time, the shift towards co-monotonicity raises the probability along the trace of the discretized transition matrix (Panel C).

These axiomatic results strengthen the case for concordance by showing it orders measures derived from different methodological traditions. Researchers often observe proxies and I now show concordance is preserved under proxy measurement.

3.5. Proxies and Measurement Error

Researchers often observe proxies—education, occupation, health status—rather than lifetime income. I show that the concordance order is maintained when using proxies.[13]

PROPOSITION 5 (Concordance of proxies). *Let a proxy E be determined by the following production function*

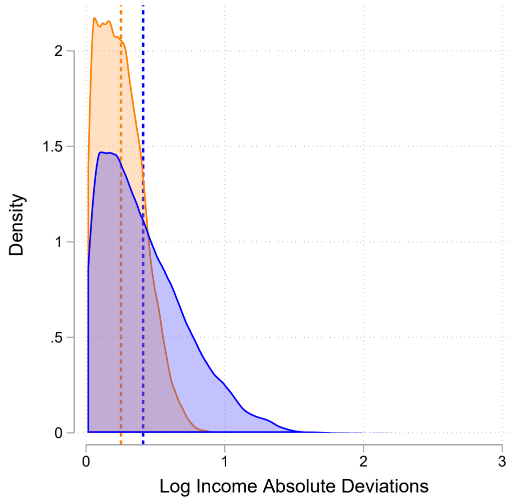
$$E^K = H^K\left(R^K, \varepsilon^K\right), \qquad E^P = H^P\left(R^P, \varepsilon^P\right).$$

Assume $H^x(\cdot,\cdot)$ *are weakly increasing function for* $x \in \{K, P\}$ *and* $(\varepsilon^K, \varepsilon^P)$ *is independent of the pair* R^K, R^P *with a joint-distribution that is the same in economy A and economy B. Then if* $C^A \succeq C^B$ *the copula of the pair* (E^K, E^P) *satisfies* $\widetilde{C}^A_E(R^K_E, R^P_E) \succeq \widetilde{C}^B_E(R^K_E, R^P_E)$*.*

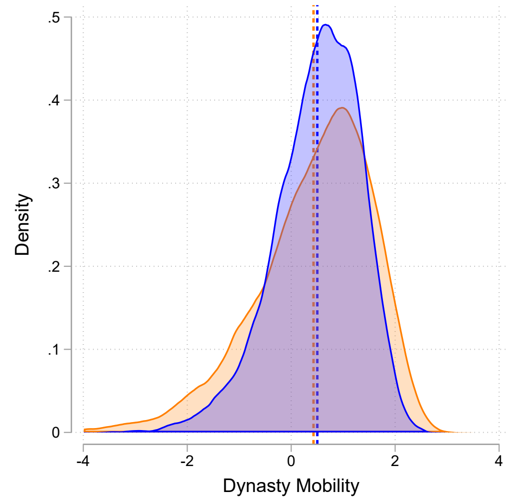
Directly modeling dependence via copulas accommodates a variety of empirically relevant proxy frameworks, e.g. in education. In particular, it captures cases where education proxies income and where latent ability drives both education and income:

[12] This is obtained by discretizing ranks into q groups. For simplicity, I consider the case where q groups are of equal size, but this is not integral to the result.

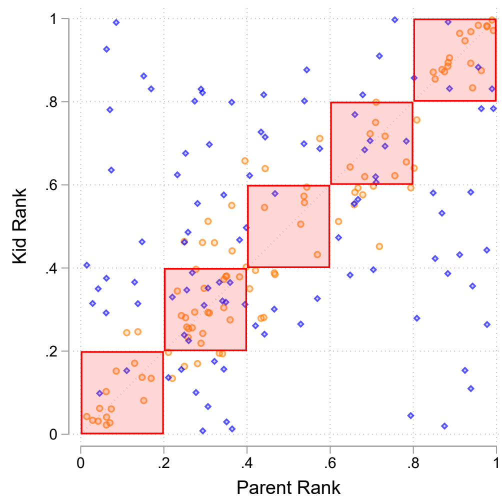
[13] Defining proxies as functions of ranks is without loss of generality under common marginals.



(A) Distribution of Fields and Ok (1996) Mobility



(B) Cowell and Flachaire (2018) Mobility



(C) Shorrocks (1978) Trace Measure

FIGURE 5. Axiomatic Measures Under Increased Concordance

Notes: Each panel shows the effect of an increase in concordance on a mobility measure. The marginal distribution of parents income and children's income are held fixed. In each panel, the correlation parameter in the elliptical copula increases and the joint-distribution of the orange distribution (circle markers) is larger than the blue joint-distribution (diamonds) in the concordance order. Panels (A) and (B) show the population distribution of individual dynasty mobility contributions as concordance changes along with mean values, corresponding to the value of the measure, in dashed vertical lines. Panel (C) shows the Shorrocks trace measure in red as regions of the rank-space using quintiles of the distribution. Panel (C) additionally shows 100 draws from the joint distribution. Results correspond to Proposition 4.

Example 1: Human capital determines incomes. Suppose increased human capital directly increases lifetime incomes (Mincer 1974) and we have $Y = H(E) + \varepsilon$. Inverting yields $E = H^{-1}(Y - \varepsilon)$. Treating ε as a vector allows for a general functional form for H and Y which can accommodate cases where the variance of the income residual increases in education as is standard in empirical work.

Example 2: Ability determines education and income. Alternatively, let a latent skill S determine both education and income: $E = H_E(S, \varepsilon)$ and $Y = H_Y(S, \varepsilon)$.

In each example the function H can differ across generations. Parent-child joint distributions, e.g. $F(Y^K, E^P)$ or $F(E^K, E^P)$, inherit the dependence structure of the copula $C(R^K, R^P)$, so concordance results extend immediately.

3.5.1. Measurement Error

Although noise attenuates measured dependence, concordance orderings are preserved under measurement error—mismeasured incomes act as proxies in the sense of Proposition 5. Thus, even though estimates like rank–rank slopes or elasticities may be biased in level, their ordering across economies is invariant to measurement error.

Corollary 1 (Concordance under mismeasurement). *Let $(\varepsilon^K, \varepsilon^P) \perp (R^K, R^P)$ denote a vector of measurement errors that are independent of true incomes or ranks with $\varepsilon^K \perp \varepsilon^P$. Reported, or observed, incomes $(\widetilde{Y}^K, \widetilde{Y}^P)$ are increasing functions of true lifetime incomes (or ranks) and the measurement error. Therefore, they satisfy, the following measurement equations*

$$\widetilde{Y}^K = G_k(R^K, \varepsilon^K), \qquad \widetilde{Y}^P = G_p(R^P, \varepsilon^P),$$

for weakly increasing functions G_K and G_P.

Holding measurement equations and the distribution of measurement errors constant (i.e. the noise structure is the same in A and B), if $C^A \succeq C^B$ then $\widetilde{C}^A \succeq \widetilde{C}^B$, where $\widetilde{C}(u, v)$ denotes the copula for the reported or observed outcomes $(\widetilde{Y}^K, \widetilde{Y}^P)$.

This result does not imply rankings are invariant to changes in measurement error. Rather, if two economies are measured with the same error structure, their concordance ranking is preserved.

Example 1: Classical measurement error in incomes. When observations are noisy measures of true lifetime income, this gives the following measurement equations satisfying the assumption in Corollary 1:

$$\widetilde{Y}^K = F_K^{-1}(R^K) + \varepsilon^K, \qquad \widetilde{Y}^P = F_P^{-1}(R^P) + \varepsilon^P.$$

Example 2: Measurement error in ranks. When ranks are calculated on a noisy measure of true incomes (as in Chetverikov and Wilhelm 2023), this can be expressed as (satisfying Corollary 1):

$$\widetilde{R}^K = F_{\widetilde{K}}\left(F_K^{-1}(R^K) + \varepsilon^K\right), \qquad \widetilde{R}^P = F_{\widetilde{P}}\left(F_P^{-1}(R^P) + \varepsilon^P\right),$$

with CDFs for the noisy income measures $F_{\widetilde{K}}(\widetilde{Y}^K)$ and $F_{\widetilde{P}}(\widetilde{Y}^P)$.

Example 3: Life Cycle Biases. Systematic variation in the link between current and life-time earnings generates bias in estimates of the intergenerational elasticity (Jenkins 1987; Nybom and Stuhler 2016), e.g.

$$\widetilde{Y}^K = G_K\left(F_K^{-1}(R^K) + \mu_{\varepsilon^K} + \sigma_{\varepsilon^K}\varepsilon^K\right), \qquad \widetilde{Y}^P = G_P\left(F_P^{-1}(R^P) + \mu_{\varepsilon^P} + \sigma_{\varepsilon^P}\varepsilon^P\right),$$

which also satisfy the restrictions in Corollary 1.

Thus, measurement error may bias estimates but does not affect concordance orderings: if true ranks satisfy concordance, so do mismeasured ranks. Similar points are made by Bhattacharya and Mazumder (2011) in the context of transition probabilities and Kitagawa, Nybom, and Stuhler (2018) in the context of rank-rank correlations, who also derive results for generalised life-cycle bias measurement equations (Example 3 above). Importantly, assuming observed measures are monotone functions of true earnings does not rule out non-classical measurement error (in the spirit of Bound et al. 1994).

3.6. Summary

Table 1 synthesizes these results, organizing mobility measures by methodological approach and confirming that concordance orders them all: regression coefficients, transition probabilities, and axiomatic indices move together when dependence changes. This coherence remains robust to mass points, measurement error, and the use of proxies like education or occupation.

TABLE 1. Properties of Exchange Mobility Measures & the Concordance Order

	Properties		Concordance Ordering				
	Copula only?	Distribution of what?	Increasing?	Mass points?	Fixed ranks?	Proxies or measurement error?	Common marginals only?
Regression measures							
IGE	No	Log income	Yes	Yes	N/A	Yes	Yes
Rank–rank correlation	Yes	Ranks	Yes	Yes	Yes	Yes	No
CER	Yes	Ranks	Yes[a]	Parametric Only	Yes	Yes	No
Transition matrix measures							
TP	Yes	Ranks	Yes	Yes	Yes	Yes	No
URM/DRM	Yes	Ranks	Yes	No	Yes	Yes	No
Axiomatic measures							
Shorrocks	Yes	Ranks	Yes[a]	Yes	Yes	Yes	No
Fields & Ok	No	Income levels	Yes	Yes	N/A	Yes	No[b]
D'Agostino & Dardanoni	Yes	Ranks	Yes	Yes	Yes	Yes	No
Cowell & Flachaire	User choice	Income levels or ranks	Yes	Yes	Yes	Yes	No[b]

Notes: [a] refers to results that additionally require Assumption 1. [b] denotes that common marginals are required unless the measure is evaluated using ranks or marginals also satisfy the usual stochastic order. Results that rule out atoms in column 4 also require continuous proxies or measurement error. Results summarize propositions in Sections 3 and 5. Proofs of the results are given in Appendix B.

4. When Concordance Fails: Completing the Partial Order

Sections 2 and 3 establish that concordance provides a powerful unifying framework: when two economies can be ranked by their copulas in the concordance order, all standard mobility measures agree on which is more mobile. This coherence explains why empirical studies often find consistent patterns across different metrics. However, concordance is only a *partial* order—not all joint distributions can be compared. This section examines what happens when concordance fails to rank two economies, why different mobility measures may then disagree, and how researchers might proceed in such cases.

4.1. The Limits of Concordance

When can concordance fail to rank two economies? Figure 6 illustrates this directly. Within each column, transition matrices increase in concordance. However, no concordance ordering exists *between* columns or within rows. The matrices in each row

Child's Quintile (Par. Quintile by row)

Par. Quintile	1	2	3	4	5
1	0.32	0.24	0.2	0.15	0.08
2	0.25	0.23	0.21	0.18	0.12
3	0.19	0.22	0.22	0.21	0.16
4	0.15	0.18	0.21	0.24	0.22
5	0.09	0.12	0.16	0.22	0.41

No Concordance Order

Child's Quintile (Par. Quintile by row)

Par. Quintile	1	2	3	4	5
1	0.58	0.00	0.00	0.14	0.28
2	0.00	0.86	0.00	0.00	0.14
3	0.00	0.00	1.00	0.00	0.00
4	0.14	0.00	0.00	0.86	0.00
5	0.28	0.14	0.00	0.00	0.58

Increasing Concordance

Increasing Concordance

Child's Quintile (Par. Quintile by row)

Par. Quintile	1	2	3	4	5
1	0.65	0.12	0.10	0.08	0.04
2	0.13	0.62	0.11	0.09	0.06
3	0.10	0.11	0.61	0.11	0.08
4	0.08	0.09	0.11	0.62	0.11
5	0.05	0.06	0.08	0.11	0.71

No Concordance Order

Child's Quintile (Par. Quintile by row)

Par. Quintile	1	2	3	4	5
1	0.79	0.00	0.00	0.07	0.14
2	0.00	0.93	0.00	0.00	0.07
3	0.00	0.00	1.00	0.00	0.00
4	0.07	0.00	0.00	0.93	0.00
5	0.14	0.07	0.00	0.00	0.79

FIGURE 6. An Example of Concordance as a Partial Order

Notes: The first row is adapted from an example in Berman (2022). Each row produces numerically equivalent values for the rank correlation.

produce numerically identical rank correlations (0.32 and 0.65, respectively), yet differ substantially in their transition probabilities or mass along the trace. This shows rank correlation alone does not determine the entire copula when concordance fails.[14]

4.2. Why Mobility Measures Disagree

When concordance fails to order two distributions, why might different mobility measures produce conflicting rankings? The answer lies in how each measure weights different regions of the joint distribution. Any mobility measure can be expressed as a weighted integral of the joint density:

$$M_M = \int\int \phi_M\left(Y^K, Y^P\right) f\left(Y^K, Y^P\right) dY^K dY^P, \tag{5}$$

where $\phi_M\left(Y^K, Y^P\right)$ is a mobility-measure specific intensity assigning importance to parent-child income pairs. When comparing economies A and B with different copulas but common marginals, the difference in any twice-differentiable measure can be

[14] Berman (2022) argues copulas in the left-column are empirically plausible (comparing to estimates in Jäntti et al. 2006, and Chetty et al. 2017), but rejects the right-column "all-or-nothing" dynamics. This reinforces the sense that the concordance restriction is implicit in applied work.

decomposed as:

$$M_M^A - M_M^B = \int_0^1 \int_0^1 \underbrace{\frac{\phi_{M,12}\left(F_K^{-1}(R^K), F_P^{-1}(R^P)\right)}{f_K(F_K^{-1}(R^K))\, f_P(F_P^{-1}(R^P))}}_{W_M(R^K, R^P)} \left[C^A(R^K, R^P) - C^B(R^K, R^P)\right] dR^K\, dR^P. \quad (6)$$

Different measures place different weights on the same changes in the copula. The intergenerational elasticity uses intensity, $\phi_M\left(Y^K, Y^P\right) = (Y^K - \overline{Y}^K)(Y^P - \overline{Y}^P)$, which emphasizes changes at the tails of the distribution—deviations from the mean among high- and low-income families matter most. The Cowell and Flachaire (2018) measure uses intensities proportional to $\phi^M\left(Y^K, Y^P\right) \propto (Y^K/\overline{Y}^K)^\alpha (Y^P/\overline{Y}^P)^{1-\alpha}$. Depending on the degree of supermodularity, α, it may emphasize extreme directional transitions more than those near the mean. Transition probability measures have indicator-function intensities—changes in density outside specific regions have *no* effect on the measure.

When concordance orders economies, $C^A(R^K, R^P) - C^B(R^K, R^P) \geq 0$ everywhere in equation (6), so all measures agree despite different weights. When concordance fails, the difference in copulas changes sign across regions. In this case, one measure may place high weight on regions where $C^A(R^K, R^P) > C^B(R^K, R^P)$ while another emphasizes regions where $C^A(R^K, R^P) < C^B(R^K, R^P)$, producing conficting rankings. The choice of measure then embeds an implicit ethical judgment over what constitutes desirable mobility (Fields and Ok 1996; Jäntti and Jenkins 2015; Ray and Genicot 2023) similar to selecting a social welfare function when Lorenz curves intersect (Atkinson et al. 1970).

4.2.1. Disagreement is Generic

Inspecting equation (6) shows that it is always feasible to construct two measures, no matter how arbitrary, that disagree on the ranking of mobility between economies that cannot be ranked by concordance. However, concordance provides a much sharper characterization of disagreement among widely used measures.

First, concordance is not just sufficient for agreement—it is necessary. If concordance fails, then by Proposition 3 there exist thresholds τ^k and τ^p for which the corresponding transition probability disagree.[15] In other words, concordance is necessary and sufficient for transition probabilities to order mobility consistently at all cutpoints the data can support.

[15] The proof of this is immediate from Proposition 3 which characterizes an *if and only if* condition on transition probabilities and concordance.

Second, even for pre-specified discretizations (e.g., quintile, decile, or ventile transition matrices) concordance violations manifest in ranking reversals among standard empirical objects.

Corollary 2 (Ranking Reversals at Standard Discretizations). *Let C^A and C^B be copulas such that $\Delta C \equiv C^A - C^B$ changes sign. Fix two reporting grids*

$$\Pi_1 = \{0 = t_0^1 < t_1^1 < \cdots < t_K^1 = 1\}, \quad \Pi_2 = \{0 = t_0^2 < t_1^2 < \cdots < t_J^2 = 1\}$$

e.g., quintiles and deciles. Define cells $R_{ij}^1 = [t_{i-1}^1, t_i^1] \times [t_{j-1}^1, t_j^1]$ and $R_{kl}^2 = [t_{k-1}^2, t_k^2] \times [t_{l-1}^2, t_l^2]$.

If there exist cells where:

i. ΔC *maintains constant sign on each cell, and*

ii. $\int\int_{R_{i,j}^1} \Delta C(u,v)dudv > 0$*, but* $\int\int_{R_{k,l}^2} \Delta C(u,v)dudv < 0$*,*

then transition probabilities reverse rankings: Grid Π_1 ranks A as less mobile, but Π_2 ranks A as more mobile. When both detecting cells lie on the diagonal ($i = j$ and $k = l$), the Shorrocks trace measure reverses rankings between the two discretizations.

Implications. When concordance fails, applied researchers using different discretizations can reach opposite conclusions about which economy is more mobile—even when analyzing identical data. A researcher using quintile transition matrices may conclude economy A has lower mobility, while another using decile matrices concludes A has higher mobility. This is not measurement error, but reflects genuine ambiguity in how to weight regions where parent-child dependence differs in opposite directions.

When concordance holds, all standard measures agree. When it fails, some standard measures must disagree. Researchers cannot avoid these violations by restricting attention to "simpler" or more "transparent"measures: Corollary 2 shows disagreement appears in the transition matrices routinely published in empirical work.

4.3. Approximate Concordance

Before imposing additional structure to complete the ordering, we note that concordance results remain approximately valid even under small violations. This matters empirically because real-world copulas may nearly—but not exactly—satisfy concordance due to sampling variation or genuine but minor departures.

PROPOSITION 6 (Continuity and Approximate Concordance). *Let M be any real-valued function of the copula C (with fixed marginals) such that*

i. *(Concordance Monotonicity) for any copulas C', C'', if $C' \succeq C''$ then $M(C') \leq M(C'')$*

ii. *(Uniform Continuity) M is uniformly continuous with respect to the copula sup-norm $||C' - C''||_\infty \equiv \sup_{(u,v)\in[0,1]^2}$.*

If there exists a copula $C^\dagger$ such that $C^A \succeq C^\dagger$ and $||C^B - C^\dagger||_\infty \leq \epsilon$ (i.e., C^B is obtained as an ϵ-perturbation of a copula $C^\dagger$ that is dominated by C^A in the concordance order), then

$$M(C^A) \leq M(C^B) + \omega_M(\epsilon)\,, \tag{7}$$

where $\omega_M(\epsilon) \equiv \sup\{|M(C') - M(C'')| : ||C' - C''||_\infty \leq \epsilon\}$ is the measure M's modulus of continuity. In particular, if M is Lipschitz continuous under the copula sup-norm ($||\cdot||_\infty$) with constant L_M, then

$$M(C^A) \leq M(C^B) + L_M\epsilon\,. \tag{8}$$

This result shows the concordance ordering is not a knife-edge property. When two economies are close to satisfying concordance–in the sense that one copula nearly dominates the other after a small perturbation—mobility measures remain approximately ordered. This bound tightens as $\epsilon \longrightarrow 0$ (exact concordance) and depends on the measure's sensitivty to copula changes.[16] Approximate concordance expands empirical relevance by allowing researchers to test whether copulas satisfy concordance within estimation error. This accommodates realistic settings where economies exhibit small departures due to institutional details or demographics while mobility measures still move together.

4.3.1. Robust Measure Choice Under Approximate Concordance

A similar approach characterize the sensitivity of measures to these deviations. Recall from equation (6) that any twice-differentiable measure M satisfies[17]

$$M_M^A - M_M^B = \int_0^1 \int_0^1 W_M(R^K, R^P)\left[C^A(R^K, R^P) - C^B(R^K, R^P)\right] dR^K\, dR^P, \tag{9}$$

[16]Moreover, for many measures the Lipschitz constant L_M is explicit or can be calculated using the Hardy-Krause variation.

[17]Analogous results apply to discrete cases, but the differentiable case simplifies exposition.

where W_M is the measure specific weighting-kernel with Lipschitz constant:

$$L_M = |W_M|_1 \equiv \int_0^1 \int_0^1 |W_M(u, v)| du\, dv\,. \tag{10}$$

COROLLARY 3 (Optimal Robust Measure). *If the true copula C satisfies* $||C - C^\dagger||_\infty \leq \epsilon$*, then* $M(C) \in [M(C^\dagger) - L_M\epsilon, M(C^\dagger) + L_M\epsilon]$*. Among concordance-monotone measures, an* ϵ*-optimal choice of measure minimizes* L_M*.*

Interpretation. Measures concentrating weight in specific regions (e.g., the intergenerational elasticity or local conditional mobility), have larger L_M and greater sensitivity to concordance violations. Measures spreading weight uniformly (e.g., Shorrocks index, coarse transitions or rank-rank slopes)—have smaller L_M and are robust to approximate concordance. This explains why rank-based measures exhibit tighter co-movement in empirical work (Deutscher and Mazumder 2023, Table 3).

4.4. Completing the Concordance Order

When concordance definitively fails—copulas exhibit substantial offsetting changes across regions—researchers face a choice. I discuss three principled approaches.

Restricting the comparison set. The first approach explicitly limits attention to economies that can be ranked by concordance. Many parametric copula families naturally induce concordance orderings as dependence parameters vary (Joe 1997, ch 4). Chapter 4. This directly rules out pathological shifts in the joint distribution in Figure 6 and retains only joint distributions identified as empirically implausible in the prior literature. The cost is loss of generality: researchers cannot compare arbitrary pairs of economies without additional modeling assumptions.

Axiomatic completion. The second approach constructs an explicit tie-breaking rule that extends concordance to a complete order. D'Agostino and Dardanoni (2009) provide axiomatic foundations for such completions in the context of discrete transition matrices. Their approach prioritizes rank distance requiring mobility increases with the magnitude of rank changes, regardless of where in the distribution they occur. This yields an intuitive minimum-distance measure.

Definition 3 (A Minimum Distance Completion of the Concordance Order). *For any empirical copula $C(u, v)$, construct the following summary statistic by mixing between the independent copula, $C^{\perp}_{uv} = uv$, and the Hoeffding upper bound, $M(u, v) = min(u, v)$:*

$$M_\lambda = \lambda^\star \in argmin_{\lambda \in [0,1]} \int_0^1 \int_0^1 \left[C(u, v) - ((1-\lambda)\, M(u, v) + \lambda\, C^{\perp}_{uv})\right]^2 du\, dv. \tag{11}$$

Intuitively, this completion asks: what mixture of perfect immobility (Hoeffding bound M) and independence ($C^{\perp}$) best approximates observed mobility? The optimal weight $\lambda^\star$ on independence provides the mobility index—higher values indicate more mobility.[18] The completion has several attractive properties. First, it reduces to pure concordance comparisons when those exist. Second, it connects to the approximate concordance results: when empirical copulas are well-approximated by the mixture in (11), the completion respects the approximate order they induce. Third, it has been implemented in related contexts: Fernández and Rogerson (2001) and Abbott et al. (2019) use an identical minimum-distance estimator to measure marital sorting.

Welfare-Consistent Completion. An alternative approach incorporates welfare judgments: a planner ranks economies based on distributional preferences. I show that, restricting the planner to concordance-respecting preferences, planner preferences provides a principled completion.

Reinterpreting equation (5), the planner's choice of measure implicitly assigns welfare weights $\phi_M\left(Y^K, Y^P\right)$. When $\phi_M\left(Y^K, Y^P\right)$ is submodular, the planner has a preference for a decrease in concordance. This completion respects concordance rankings while breaking ties via planner preferences. It accommodates utilitarian welfare with concave utility, opportunity-equality objectives minimizing rank correlation, and poverty-focused weights concentrating on low-rank dynasties (Atkinson and Bourguignon 1982).

4.5. Connection to Inequality Measurement

The relationship between concordance and mobility measurement parallels a foundational result in inequality measurement. Second-order stochastic dominance provides a partial ordering over income distributions: when one distribution

[18] It is easy to show using the usual first order condition that the value of this mobility measure is given by $M_\lambda = 1 - \int_0^1\int_0^1 \left(C(u, v) - u\, v\right)\left(\min(u, v) - u\, v\right) du\, dv \Big/ \int_0^1\int_0^1 \left(\min(u, v) - u\, v\right)^2 du\, dv$.

dominates another, all inequality measures satisfying transfer-sensitivity (e.g., Gini coefficients) agree on rankings (Atkinson et al. 1970). Concordance plays an analogous role for mobility.[19]

Second-order stochastic dominance varies inequality at fixed means; concordance varies dependence at fixed marginals. Elementary Pigou-Dalton transfers progressively equalize distributions; concordance-increasing exchanges (Definition 2) progressively align ranks across generations. Just as Lorenz curve domination ensures all transfer-sensitive inequality indices agree, concordance rotation of conditional expected rank curves (Figure 3) ensures all exchange-mobility measures agree. When stochastic dominance fails—Lorenz curves cross—a complete welfare ranking requires specifying a social welfare function. When concordance fails, completing the mobility ordering requires analogous normative choices about which distributional features matter most.

This parallel extends beyond methodology to interpretation. Inequality measurement recognizes that summary statistics embed value judgments; the choice between Gini, Theil, or variance reflects different sensitivities to transfers at different income levels. This framework makes explicit that mobility measurement involves similar choices. Concordance delineates when such choices are irrelevant (because all measures agree) from when they determine conclusions. This transparency helps researchers understand when disagreement across measures reflects definitional differences rather than economic substance.

Section 5 next examines how changing marginal distributions across time and place affect these conclusions.

5. Comparisons Across Time and Place: When Marginals Differ

Previous results assumed common marginals to isolate variation in only the dependence structure. This corresponds to comparing hypothetical variants of a single economy. Comparisons across space and time, such as asking whether U.S. is more mobile today than it was in 1965 (Chetty et al. 2017) or whether the U.S. is more mobile than Canada

[19] A similar connection was first proposed by Dardanoni (1993) who coins a *Dynamic* Pigou-Dalton principle, although earlier work connects elementary operations in Definition 2 to concepts of mobility.Atkinson and Bourguignon (1982) highlight precisely these switches in the context of multi-dimensional inequality. This connects the results on measuring mobility across generations to measuring social welfare. When social welfare is utilitarian and sub-modular at the level of parent-child incomes, increased concordance lowers both mobility and social welfare.

(Corak 2020), may involve different marginal income distributions. I now examine whether the results survive this generalization.

As long as copulas are ordered, all rank-based measures—which isolate 'pure' dependence independently of marginal distributions—are ordered even without common income distributions. Proxy-based indices mapping ranks (e.g., social classes to occupations) are similarly robust.[20] In contrast, however, proxies that depend directly on the levels of income are not ordered.[21]

This robustness does not extend to level- or log-based mobility measures.[22] Consider an economy in which parent and child incomes are distributed log-normally with

$$\begin{pmatrix} \ln Y^K \\ \ln Y^P \end{pmatrix} \sim N\left[\begin{pmatrix} \mu_P \\ \mu_K \end{pmatrix}, \begin{pmatrix} \sigma_K^2 & \rho \\ \rho & \sigma_P^2 \end{pmatrix}\right],$$

where ρ controls the dependence between generations and orders the underlying copula in concordance. In this special case, the rank-rank correlation is given by $Corr(R^K, R^P) = 6/\pi \arcsin(\rho/2) \approx \rho$ (Kruskal 1958) so that both the intergenerational correlation and the rank-rank correlation are almost identical. Thus, they have the same ordering over economies with different values of ρ. However, even in this case the ordering of the intergenerational elasticity, $\beta = \rho\sigma_K/\sigma_P$, can reverse even as ρ increases because it depends on changes in the marginal distributions of Y^K and Y^P through their relative dispersion as well as their dependence. More generally, differences in the shape and scale of incomes can tilt the measurement of mobility. Column 6 of Table 1 summarizes these limits, highlighting where concordance ordering holds and where it does not.

Section 3 established when increased concordance orders changes in mobility measures. A remaining question is whether concordance captures economically meaningful variation or merely statistical dependence. Section 6 provides an explicit economic microfoundation.

[20] This implies rank based alternative versions of measures are ordered. For example, the Fields and Ok (1996) measure computed using ranks which corresponds to the Bartholomew (1973) average jump measure, a seminal contribution in quantitative sociology.

[21] The assumptions that proxies depend on ranks, and not levels, can be tested in auxiliary datasets without information on two generations.

[22] Under the stronger assumption of stochastic dominance of the marginal distributions, it is possible to obtain a general ranking for all super-modular measures (see Meyer and Strulovici 2013, for an example of this decomposition argument). This approach is particularly relevant for exercises that employ "copula and marginal" approximations to estimate absolute mobility. Additionally, a variation on the approximate concordance order obtained in the previous section applies. When marginals are not identical, but are sufficiently close in the sense of bounded differences, continuity results apply an ordering over measures analogously to results in Proposition 6.

6. A Structural Interpretation of Concordance

To give concordance an economic interpretation, I show how it arises naturally in Becker and Tomes (1979)- Loury (1981) models of human capital investment and intergenerational mobility, demonstrating that it captures standard economic mechanisms rather than statistical artifacts.[23]

Consider a standard dynastic model. A family consists of a parent and child indexed by P and K. Parents value their own consumption and their child's consumption, with altruism parameter $\delta = 1/1+r$. They choose to consume, C, and invest, I, in the future human capital of their child. The human capital of their child depends on investment and the direct transmission of the parent's human capital, H^P, which are aggregated through a weakly decreasing returns to scale production function f. It also depends on a term capturing luck, θ. They solve the following maximisation problem:

$$\max_{C^P, I} U\left(C^P\right) + \delta \mathbb{E}_\theta \left[U\left(C^K\right)\right] \tag{12}$$

subject to

$$C^P = Y^P - I \tag{13}$$

$$C^K = Y^K = W(H^K) \tag{14}$$

$$H^K = \theta f(I, H^P) \tag{15}$$

When the production function is Cobb-Douglas and wages are linear in human capital this produces the standard intergenerational earnings elasticity specification. As in Loury (1981), the ability shock θ is realised after investments are made and is independent of familiy choices and state variables. Consequently, θ becomes the error-term in the reduced form intergenerational earnings elasticity specification.

I now show that policy or technology changes that expand investment—making investment $I^\star(Y_P)$ higher for all parent incomes—necessarily increase concordance. Therefore, it also shares a common microfoundation with the intergenerational elasticity, e.g. Solon (1992). This provides an economic interpretation: more mobility reflects weaker investment responses to parental resources.

For tractability, I additionally assume that the income of each child is an increasing

[23] I abstract from several empirically important margins including parental beliefs, dynamic complementarity, multidimensional skills, and borrowing constraints (see e.g., Cunha and Heckman 2007; Caucutt and Lochner 2020; Attanasio, Meghir, and Nix 2020; Caucutt et al. 2020; Moschini 2023).

function of either human capital or their rank in the human capital distribution.[24] Given any equilibrium distribution of children's human capital this can be justified through a number of different microfoundations: a piece-rate in efficiency units; the assignment of heterogenous workers to heterogenous firms when production is supermodular (Becker 1973; Sattinger 1975; Rosen 1981; Shimer and Smith 2000); rank-order tournaments (Lazear and Rosen 1981); or a hierarchical job-assignment model (Costrell and Loury 2004). Importantly, it is analytically convenient to separate the role of sorting or intergenerational dependence from the marginal distribution of child incomes. As it is ultimately the rank of the child's human capital that matters, this might reasonably be termed a model of economic status production.

I focus on increased concordance generated through the endogenous investment decision. Before proving the general result, note that many economic forces increase investment uniformly: higher returns to skill (wage schedule W), better learning technology (production function $f(\cdot,\cdot)$), or greater patience (discount factor δ). Lemma 2 shows any change that uniformly increases investment increases concordance.

Lemma 2 (Concordance and an outward expansion of investment). *Assume that the human capital production function $f(\cdot,\cdot)$ is weakly increasing and supermodular in both arguments. Then, holding the distribution of parental incomes fixed, any two economies A and B that produce the following pointwise relation on endogenous investment decisions $I^{\star}$,*

$$I^{\star A}\left(Y^P\right) \geq I^{\star B}\left(Y^P\right) \quad \forall\, Y^P, \tag{16}$$

also produce the following concordance order on parent-child incomes and their copulas:

$$A \succeq B \quad \textit{equivalently} \quad C^A \succeq C^B. \tag{17}$$

The following corollary links this directly to interpretable model primitives.[25]

Corollary 4 (Mechanisms producing increased concordance). *The following mechanisms all imply a pointwise order on investment policy functions, $I^{\star A}\left(Y^P\right) \geq I^{\star B}\left(Y^P\right)$ $\forall Y^P$, and, thus, a concordance order:*

[24] This is a small abuse of notation. In comparative statics exercises this does not hold the marginal distribution of child incomes constant, but still allows economies to be ranked by the concordance of their copulas. It is trivial to see that this holds when we impose common marginals. More generally, this sets the model in a general equilibrium context (following Heckman, Lochner, and Taber 1998 or Lee and Wolpin 2006 for example) by allowing wages to adapt to changes in the supply and composition of educated workers without fully specifying general equilibrium forces.

[25] In Appendix D, I prove that a broad class of policy or parameter shifts, each raising the marginal return to investment, expand the investment policy function and tighten parent–child dependence.

i. An increase in the marginal product of investment, $f_I^A(I, H^P) \geq f_I^B(I, H^P)\ \forall H^P, I$, *or returns to human capital,* $W^A(H^K) \geq W^B(H^K)\ \forall H^K$;

ii. An investment subsidy or parental-income targeted investment intervention;

iii. An increase in the discount factor, $\delta^A \geq \delta^B$.

In economic terms, amplifying the motive to invest increases the equilibrium contribution of investment.[26] Thus, concordance is not just a statistical artifact, but is instead consistent with the optimizing behaviour of economic agents.

However, not all economically relevant changes take this form. Concordance can fail to order economies when interventions affect parent-child dependence heterogeneously. The following empirically salient example (Jackson, Johnson, and Persico 2016) illustrates this. When resources are reallocated toward poor districts and away from rich districts, this effectively lowers investment at high parental income and raises it at low incomes, thus $I^{\star A}\left(Y^P\right) - I^{\star B}\left(Y^P\right)$ changes sign. This generate regions where $\Delta C > 0$ and regions where $\Delta C < 0$, violating the concordance ordering and producing measure-dependent mobility rankings.

7. Revisiting Intergenerational Mobility Over the 20th Century

Having established theoretical conditions under which mobility measures agree, I now test these predictions using U.S. data across the 20th century.

To link the theoretical results explored above to the study of intergenerational mobility, I construct cohort-specific mobility estimates for U.S. males born between 1910 and 1980. Following Jácome, Kuziemko, and Naidu (2025), I pool all available surveys that report respondents' current family income together with race, father's occupation, and region of birth or childhood and impute parental income from auxiliary data sources (primarily the Census) using race, education, and region.[27] This yields a

[26] This is closer to a 'luck' interpretation of θ. In more recent work, e.g. Cunha, Heckman, and Schennach (2010); Caucutt and Lochner (2020), this θ is known at the time of investment and is the child's ability to learn or the parent's ability to teach. Under this alternative timing, the pointwise ordering of investment below instead holds for all (θ, Y^P) pairs and the expectation is degenerate. It is also possible to incorporate stochastic wages given human capital, although I omit this for parsimony. As Lochner and Park (2024) emphasize, in this case intergenerational income mobility will differ from intergenerational skill mobility.

[27] For linear estimating equations this is a Two-Sample Instrumental Variable design. Like many studies of mobility in a historical context (e.g., Collins and Wanamaker 2022; Ward 2023), this application uses both self-reported incomes and a proxy approach to construct linked parent-child outcomes. The theoretical characterisation of mobility measures shows that results are robust to using proxies or the presence of measurement error.

repeated cross-section that is nationally representative and consistent over time. This allows me to document trends in intergenerational mobility measures over the 20th century (see also, Davis and Mazumder 2024).

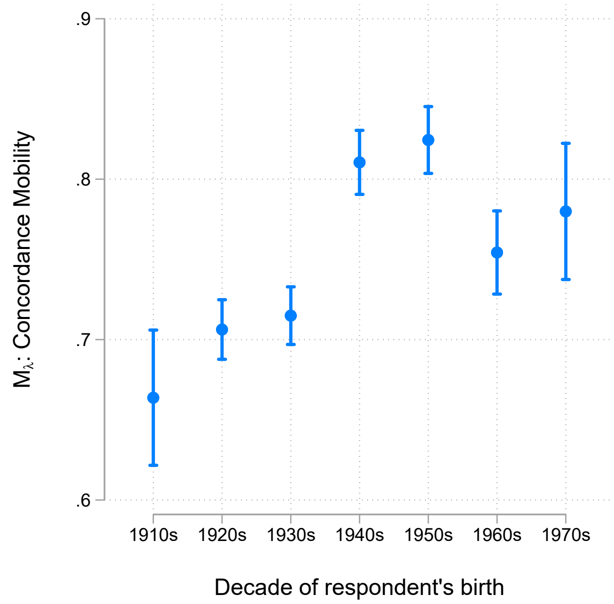


FIGURE 7. Concordance Mobility Over the 20$^{\text{th}}$ Century

Notes: Author's estimates from 15 combined US data sources (see text, Jácome, Kuziemko, and Naidu 2025 and Appendix E for details) by birth decade for respondents ages 30–50. Parental income predicted using family income conditional on father's occupation, race, and region (South vs. elsewhere) from auxiliary data (often the census) as close as possible to the respondents' tenth birthday (see Jácome, Kuziemko, and Naidu 2025 sec. III.B for more details). Sample weights are used and I reweight each birth cohort (i.e., decade) so that they have representative race$\times$sex shares. Estimate of concordance mobility measure in (11) and 95% confidence interval from 400 bootstrap replications.

I restrict the sample to fathers and sons aged between 30 and 50 (to limit life-cycle bias) and begin by documenting trends in the concordance measure of economic mobility using the completion of the partial order in equation (11). Figure 7 shows how the (dis-)concordance measure (Equation 11), the mixture-weight between co-monotone and independent copulas, evolves over the 20th century. Mobility increases from 0.66 to 0.78 over the 20th century. Mobility increases in the first half of the century before declining for the 1960s birth cohort. The point estimates are precisely estimated and the growth in intergenerational mobility is statistically significant. Notably, mobility has a wave-like pattern and does not rise monotonically. Theory predicts other rank-based measures should display the same pattern, which I now demonstrate.[28]

I now document trends in other intergenerational mobility measures in Figures 8 and 9: the intergenerational elasticity (IGE), the log-income correlation, and the rank-rank slope, along with directional and trace-based measures from the transition matrix, and axiomatic mobility measures. I normalize measures so that they increase with mobility (i.e., one minus persistence). Remaining details of the implementation are deferred to Appendix E.

Panel (A) of Figure 8 replicates the broad patterns in Jácome, Kuziemko, and Naidu (2025):[29] rank persistence and the income correlation fall, thus mobility rises, through the mid-20th century and then stabilise, while the IGE rises as income dispersion widens in the latter half of the century leading to a reversal of mobility. As established in Section 5, this divergence reflects the IGE's sensitivity to changing marginal dispersion. Growing income inequality in the lower tail mechanically increases the IGE even as rank-based persistence falls. Turning to measures of directional mobility, Panel (B) shows that both Downward and Upward Rank Mobility (see Proposition 3 for definition) measures trend upwards over the 20th century.

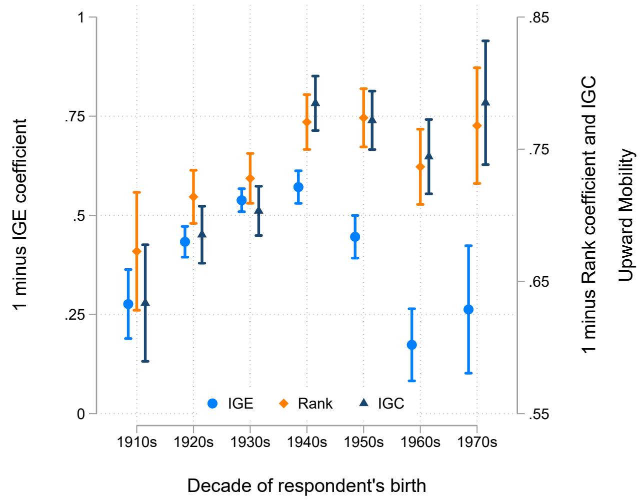
Under the concordance order, the rank-rank mobility measure in Panel (A) is a sufficient statistic for all rank-based relative mobility measures. Panels (B)-(D) of Figure 8 show that this theoretical prediction is supported by the data. The wave-like pattern of decreasing rank-persistence (Panel A) is mirrored in the estimates of increasing directional mobility. Panel (C) shows the same trends hold true for mobility measures using the diagonal of the transition matrix. Both so-called "rags-to-rags" and "riches-to-riches"[30] exhibit a trend of declining persistence. Similarly Panel (C) also shows the Shorrocks measure tracks the same wave-like pattern of persistence. Finally, Panel (D) displays the analogous off-diagonal transitional probabilities.

Figure 9 repeats the exercise for two axiomatic indices. Panel (A) plots the Exchange mobility measure of Fields and Ok; panel (B) shows the Cowell and Flachaire measure with "relative status" computed with income ranks and levels. Both rise across successive birth cohorts, echoing the upward trends in rank-based measures from Figure 8, even

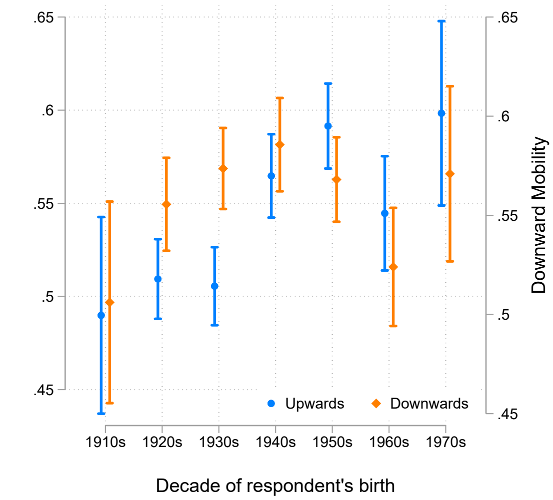
[28]This wave-like pattern—rising through mid-century, plateauing thereafter—likely reflects the interplay of expanding secondary education, compression of wage structures during the 1940s-70s, and subsequent increases in skill premia, though identifying specific mechanisms is beyond this paper's scope. See Davis and Mazumder (2024) for further discussion.

[29]This replicates results in their Figure 1. However, due to differences in the underlying datasets (see Appendix E) this is best thought of as a distinct replication rather than a reproduction. Results are qualitatively identical despite modest quantitative differences; lending support to the overall validity of their results.

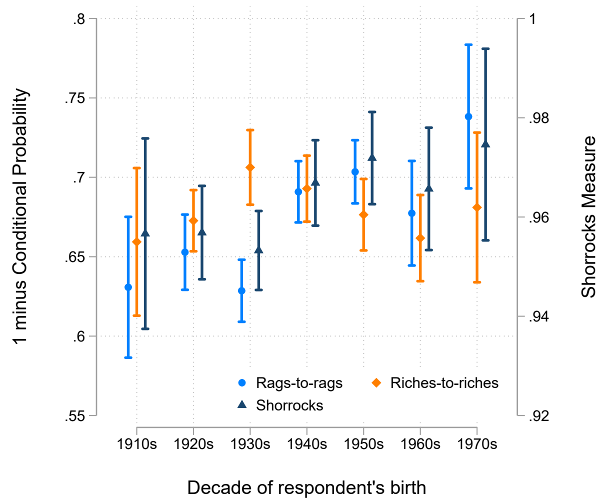
[30]Remaining in the bottom and top quintile across generations, respectively.



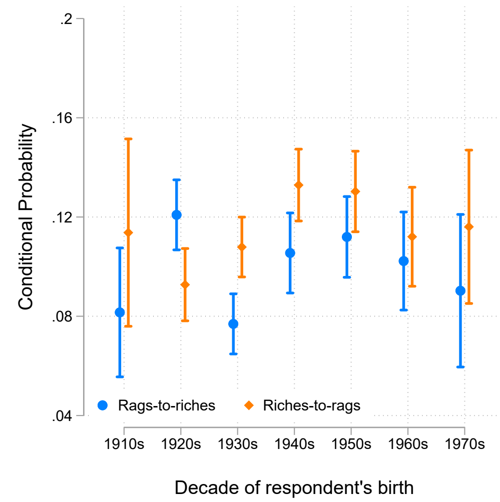
(A) Regression Estimates



(B) Directional Rank-Mobility



(C) Transition Probabilities (On-Diagonal)



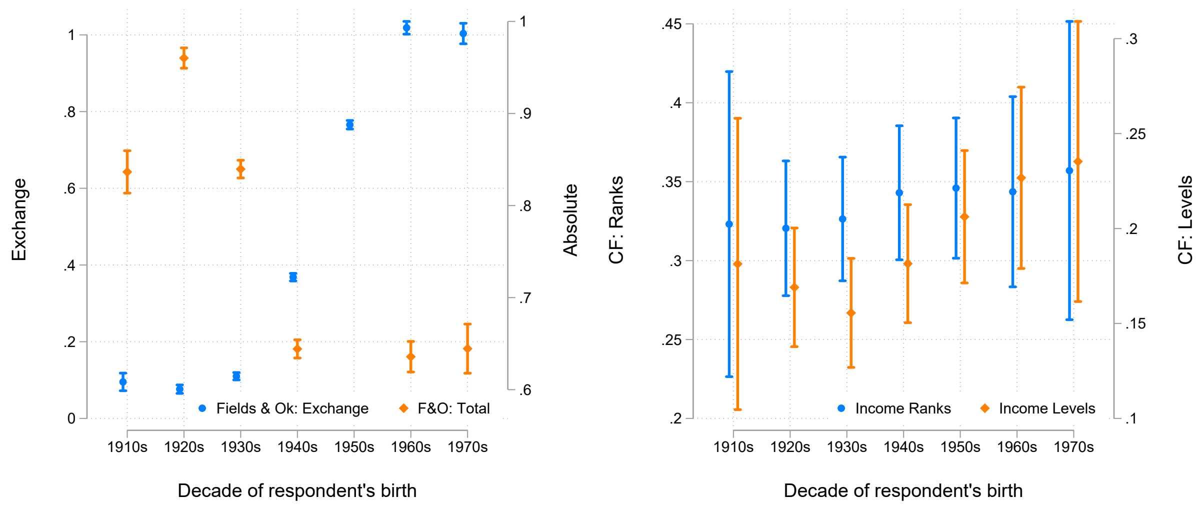
(D) Transition Probabilities (Off-Diagonal)

FIGURE 8. Trends in Intergenerational Mobility Measures

Notes: See Figure 7 for data construction details. Panel (A) reports estimates of IGE, IGC, and rank-rank slope corresponding to Proposition 1. Panel (B) reports estimates of directional rank mobility measures using quintiles ($Pr(R^K - R^P > 0.2 \mid R^P \leq 0.2)$ and $Pr(R^K - R^P < -0.2 \mid R^P > 0.8)$) in Proposition 3. Panels (C) and (D) report transition probabilities using quintiles (Proposition 3) and panel (C) additionally reports the Shorrocks Trace Measure (Proposition 4) using deciles. Panels (B)-(D) use 400 bootstrap replications for inference. Panels (A) and (C) normalise measures to increase with mobility (one minus persistence).

though the concordance order does not, by itself, determine the behaviour of these measures when marginals differ across cohorts.

By contrast, the absolute version of the Fields and Ok index, which incorporates growth in mean income and changes in inequality, tells a different story. It climbs sharply for early-century cohorts but flattens thereafter, much like the reversal in the IGE. Aggregate income growth concentrated in the right tail of the income distribution (see, e.g., Blundell et al. 2018; Piketty, Saez, and Zucman 2018; Guvenen et al. 2022) offset the gains in relative mobility, leaving absolute mobility roughly constant in the latter half of the century.



(A) Fields & Ok Measure

(B) Cowell & Flachaire Measure

FIGURE 9. Trends in Intergenerational Mobility Measures (cont.)

Notes: See Figure 7 for data construction details. Panels (A) and (B) report estimates of the Fields & Ok and Cowell & Flachaire measures (Proposition 4), respectively.

Despite imprecision in estimates of some mobility measures, results align closely with the theory. As the estimated rank-rank correlation and, therefore, the concordance falls across cohorts, the rank–rank slope falls, directional “rags-to-riches” and “riches-to-rags” probabilities increase, and the Shorrocks trace declines. This is exactly the pattern implied by higher concordance and the mobility measure from the completion of the concordance order. Absolute-mobility indices that incorporate mean growth and inequality rise early in the century but flatten thereafter, showing how shifting marginals can offset gains in relative mobility.

8. Discussion and Implications for Applied Researchers

The core insight in this paper is straightforward: when two economies or time periods can be ranked by concordance, any mobility measure—whether based on ranks, regressions, transition probabilities, or axiomatic distances—will agree on which setting is more or less mobile. In other words, immobility rises and mobility fall in tandem, across a wide array of exchange-mobility metrics if their underlying copulas are ordered by concordance.

This helps answer two important open questions in the literature on intergenerational mobility (e.g., Berman 2022 and Deutscher and Mazumder 2023). First, when is one measure sufficient? If a collection of joint distributions is ordered by concordance, then any one metric serves as a sufficient statistic for all the others. This applies ordinally, but is sufficient to rank economies in terms of more or less mobility.

Second, which measure should the applied econometrician choose? Axiomatic arguments may not suffice, as under concordance all indices deliver the same ordering and thus the order shares a common axiomatic justification. Whenever two distributions cannot be compared under concordance, selecting a particular statistic amounts to choosing a tie-breaking rule to resolve incomparable cases. When they are ordered under concordance the choice of measures is, in a sense, redundant. Without an ex ante reason to prefer one measure to another a researcher may possess the ability to choose the ordering ex post (although inference on rankings is challenging, see Mogstad et al. 2024).

Even when measures agree empirically, concordance explains why. Beyond theoretical equivalence, practical considerations favor specific measures under data constraints. Transition probabilities require only ordinal rankings and few parameters, making them robust to small samples and measurement error (Corollary 1). Chetty et al. (2014a) exploit this when constructing local mobility estimates. The IGE, conversely, requires precisely measured log incomes and is sensitive to life-cycle bias (Haider and Solon 2006) and top-coding, limiting its use with noisy or cross-country data.

When only proxies are available—e.g., education or occupation in historical studies (Ward 2023) or in developing countries (Neidhöfer, Serrano, and Gasparini 2018)—then Proposition 5 shows rank-based measures preserve concordance. Rank-rank slopes offer additional advantages: they are unit-free (aiding cross-country comparison), robust to specifications Chetty et al. (2014b), and accommodate the proxy framework directly. Under concordance, researchers can confidently select measures based on data quality rather than concern about missing substantive information.

Some measures live in more familiar units—elasticities, correlation slopes or probabilities—which can aid interpretation—however, this may also allow researchers to manipulate notions of ‘similar’ mobility. Other measures may be better suited to descriptive or causal analysis of the underlying mechanisms. For example, they may be easier to decompose into the mobility of subgroups or be better suited to mediation analysis (as used in Fagereng, Mogstad, and Rønning 2021; Bolt et al. 2021). Finally, practical data considerations often drive the choice of measure: education proxies, for instance, are widely available even when lifetime-income ranks are not. Under concordance, those proxies still yield valid comparisons of relative positions.

The concordance order is primarily a statistical concept, but it also captures economically meaningful changes in behaviour. It naturally arises as a way to characterize comparative statics in a standard model of endogenous human capital formation. This provides a novel microfoundation for the abstract concept.

Concordance also makes it possible to knit together mobility rankings across datasets that use different measures. When economies are linked by a chain of pairwise concordance comparisons—so that each economy shares at least one mobility metric with another in the chain the ordering can be propagated throughout that connected group, even if no single measure is common to all. This greatly expands the scope for cross-country and cross-period mobility comparisons while preserving a single concept of relative mobility and immobility.

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Online Appendix

Appendix A. Discussion and Proofs for Section 2

A.1. Copula Uniqueness

Sklar's Theorem (1959) states that the copula associated with a joint distribution is unique if and only if marginal distributions are continuous. When the random variables take on discrete values or are drawn from distributions with mass points, the copula is generically not uniquely determined. Instead, it is only unique on the set of ranks the marginals actually take (Thm 2.3.3, Nelsen 2006). There are infinitely many valid extensions for values outside the observed support of ranks.

Thus, even if mass points are only at the boundary of support (as in the case of censored or truncated income observations), the copula is not unique. The results in this paper do not, unless stated otherwise, rely on continuity or differentiability of the copula. However, they do implicitly assume uniqueness of the copula. Therefore, discreteness represents a problem as it leads to indeterminacy.

Addressing this indeterminacy is referred to as an *extension* of the copula. Unfortunately, extensions are not necessarily guaranteed to preserve the concordance order. Thus, the choice of extension is not without loss. There are three solutions to this problem for a given empirical copula:

(i) Use a checkerboard extension (Genest and Nešlehová 2007) which is piecewise bilinear

(ii) Consider Carley (2002) bounds on copulas

(iii) Approximate the discrete outcomes with a specific parametric copula family

Solution (iii) is the easiest solution, but also requires the strongest assumptions making it typically unappealing. Solution (ii) would replace inequalities on the copula required for the concordance order with worst-case inequalities comparing upper and lower bounds. This is extremely flexible, but can be limiting when the bounds are not sharp. Finally, solution (i) restricts the set of admissible extensions and selects a bilinear interpolation on the density. This guarantees uniqueness and, as shown by Propositions 11 and 13 in Genest and Nešlehová (2007), the checkerboard extension is the unique extension that preserves the concordance order of the original joint-distributions. Consequently, I assume that discreteness is addressed through checkerboard extension of the copula. While this is a mild assumption, equivalent results can be obtained under either alternative assumption.

A.2. Proofs

Proof of Lemma 1. Let $u1 = v1 = 0$, $u_2 = u$ and $v_2 = v$. Then the boundary conditions of the copula give

$$C(0, v) = C(u, 0) = C(0, 0) = 0 \tag{A.1}$$

for all copulas. Thus Assumption 1 is equivalent to

$$\Delta C(u, v) \geq 0, \tag{A.2}$$

which is identical to Definition 1. □

Appendix B. Proofs for Section 3

Proof of Proposition 1. The population value of β is given by $cov(\ln y_i^k, \ln y_i^p)/var(\ln y_i^p)$. The denominator is constant and the numerator is increasing relative to the concordance ordering as a direct consequence of Lemma 3 in Lehmann (1966) (see also Lemma 2, Hoeffding 1940). Similarly, the population value of ρ is increasing relative to the concordance ordering from Corollary 3.2. in Tchen (1980). □

Proof of Proposition 2. In the parametric case, the conditional expectation is given by (substituting the closed form value of α):

$$CER = 0.5 + \rho(R - 0.5) \tag{B.1}$$

which is increasing in ρ above the median ($R > 0.5$) and decreasing below ($R < 0.5$). Consequently, given ρ is increasing relative to the concordance ordering (Proposition 1), the stated inequalities follow. This does not rely on Assumption 1.

Define

$$D(v) = \int_0^1 u\, \partial_v C^A(u, v)\, du - \int_0^1 u\, \partial_v C^B(u, v)\, du.$$

Then there exists a unique $v^* \in (0, 1)$ with $D(v^*) = 0$; equivalently, $\mathbb{E}^A[\, U \mid V = v\,]$ and $\mathbb{E}^B[\, U \mid V = v\,]$ cross exactly once.

To establish this we impose an additional regularity assumption: $D(v)$ is single crossing on $[0,1]$. A stronger sufficient condition for this regularity assumption is to require the family of conditional densities satisfy a monotone-likelihood-ratio or TP_2 properties in models that admit densities or for many parametric copula family. This guarantees uniqueness of the root.

First, consider the left tail of v. We establish that $D(v) > 0$ in the neighborhood of $v = 0$. Fix $\varepsilon \in (0,1)$. Using $C^A(u,0) = C^B(u,0) = 0$ and $C^A(u,\varepsilon) \geq C^B(u,\varepsilon)$ for every u gives

$$\int_0^{\varepsilon} D(v)\, dv = \int_0^1 u\big[C_1(u,\varepsilon) - C_2(u,\varepsilon)\big]\, du > 0 \quad (\varepsilon > 0).$$

$D(0) = 0$ and D is continuous, thus some $v_0 \in (0,\varepsilon)$ satisfies $D(v_0) > 0$. Equivalently, in the neighborhood of $v = 1$ $u - C^A(u, 1-\varepsilon) \leq u - C^B(u, 1-\varepsilon)$, hence

$$\int_{1-\varepsilon}^1 D(v)\, dv < 0.$$

Thus, there exists $v_1 \in (1-\varepsilon, 1)$ with $D(v_1) < 0$. D is continuous, positive at v_0, and negative at v_1; by the Intermediate Value Theorem it has a root $v^* \in (v_0, v_1)$. Under the regularity assumption, a zero can occur *at most once*. Hence v^* is the unique crossing. □

Proof of Proposition 3. The transition probabilities can be rewritten using the conditional CDF. For the positive diagonal case we have

$$TP\left[R^K \leq \tau^k \mid R^P \leq \tau^p\right] = \frac{Pr(R^K \leq \tau^k \cap R^P \leq \tau^p)}{Pr(R^P \leq \tau^p)} = \frac{C(\tau^k, \tau^p)}{\tau^p}, \tag{B.2}$$

and

$$TP\left[R^K > \tau^k \mid R^P > \tau^p\right] = \frac{Pr(R^K > \tau^k \cap R^P > \tau^p)}{Pr(R^P > \tau^p)} = \frac{1 - \tau^p - \tau^k + C(\tau^k, \tau^p)}{1 - \tau^p}, \tag{B.3}$$

with the inequality in the proposition satisfied iff

$$C^A(\tau^k, \tau^p) \geq C^B(\tau^k, \tau^p)\ \forall\, \tau^k, \tau^p \in [0,1]\,, \tag{B.4}$$

which is identical to the definition of increased concordance.

For the off-diagonal case we have

$$TP\left[R^K > \tau^k \mid R^P \leq \tau^p\right] = 1 - TP\left[R^K \leq \tau^k \mid R^P \leq \tau^p\right], \tag{B.5}$$

and

$$TP\left[R^K \leq \tau^k \mid R^P > \tau^p\right] = 1 - TP\left[R^K \leq \tau^k \mid R^P > \tau^p\right], \tag{B.6}$$

with the stated inequality following directly from the results established in the positive diagonal case.

Turning to directional rank mobility. Fix $\tau \in (0, 1]$ and an s such that $1 - \tau \geq s \geq 0$. First, note that we can express the following conditional density through Bayes' rule

$$Pr(R^K - R^P > s \mid R^P \leq \tau) = \frac{Pr(R^K - R^P > s \cap R^P \leq \tau)}{Pr(R^P \leq \tau)} = \frac{Pr(R^K > R^P + s \cap R^P \leq \tau)}{Pr(R^P \leq \tau)}, \tag{B.7}$$

where the numerator is the probability that both conditions hold. This can be rewritten as

$$Pr(R^K > R^P + s \cap R^P \leq \tau) = \int_0^\tau Pr(R^K > t + s|R^P = t)\, dt, \tag{B.8}$$

which uses the fact that ranks are uniformly distributed and bounded between 0 and 1. The conditional density is well-defined (Theorem 2.2.7., Nelsen 2006).

As $Pr(R^P \leq \tau) = \tau$ and is constant across copulas, the stated inequality is identical to establishing the inequality pointwise on the integrand in equation (B.8).

Let

$$\begin{aligned} g_h(x) \equiv Pr(R^K > x + s \mid x < R^P \leq x + h) &= 1 - Pr(R^K \leq x + s \mid x < R^P \leq x + h) \\ &= 1 - \frac{Pr(R^K \leq x + s\,, x < R^P \leq x + h)}{h}, \end{aligned}$$

where the alternative definition is a direct application of Bayes' rule. The integrand in equation (B.8) can then be expressed as

$$Pr(R^K > t + s|R^P = t) = \lim_{h \downarrow 0} g_h(t), \tag{B.9}$$

where the function $g_h(\cdot)$ is continuous in h by uniform continuity of the copula (corollary 2.2.6., Nelsen 2006). Note that $C(u, v + h) - C(u, v) = Pr(R^K \leq u\,, v < R^P \leq v + h)$. We can then express $g_h(x)$ in terms of copulas as

$$g_h(x) = 1 - \frac{C(x + s, x + h) - C(x + s, x)}{h} = 1 - \frac{Pr(R^K \leq x + s\,, X < R^P \leq x + h)}{h}. \tag{B.10}$$

Assumption 1 guarantees $\Delta C(u, v+h) \geq \Delta C(u, v)$ and

$$C^A(u, v+h) - C^A(u, v) \geq C^B(u, v+h) - C^B(u, v) \tag{B.11}$$

$$\longrightarrow \frac{C^A(u, v+h) - C^A(u, v)}{h} \geq \frac{C^B(u, v+h) - C^B(u, v)}{h}. \tag{B.12}$$

Substituting (B.12) into (B.10), establishes

$$g^A(x) \leq g^B(x) \longrightarrow \lim_{h \downarrow 0} g^A(x) \leq \lim_{h \downarrow 0} g^B(x) \longrightarrow \int_0^\tau \lim_{h \downarrow 0} g^A(t)\, dt \leq \int_0^\tau \lim_{h \downarrow 0} g^B(t)\, dt, \tag{B.13}$$

thus $Pr^A(R^K > R^P + s \cap R^P \leq \tau) \leq Pr^B(R^K > R^P + s \cap R^P \leq \tau)$. □

Proof of Proposition 4. Fields and Ok Measure. The aggregate income movement per capita is equivalent to

$$\int \int \psi\left(Y^K, Y^P\right) f\left(Y^K, Y^P\right) d\,Y^K d\,Y^P \quad \text{where } \psi\,(x, y) = |x - y|, \tag{B.14}$$

where the function $\psi(x, y)$ is a convex function of the difference. The result then follows directly as an application of Theorem 9.A.18 in Shaked and Shanthikumar (2007) or Corollary 2.3 (and example 2) in Tchen (1980). Expected absolute deviations across generations are decreasing in the concordance order. The proof in the log-case (Fields and Ok 1999) is identical. The Fields and Ok (1996, 1999) measures can be decomposed into structural mobility (an aggregate growth term),

$$\int |Y^K| f^k(Y^K) d\,Y^K - |Y^P| f^P\left(Y^P\right) d\,Y^P, \tag{B.15}$$

which is identical under C^A and C^B with constant marginals as well as a transfer or exchange term

$$2 \times \int \int_{(Y^K, Y^P) \in S} |Y^K - Y^P|\, f(Y^K, Y^P)\, dY^K\, dY^P, \tag{B.16}$$

where the set S selects individuals who are winners or losers. These are dynasties that experience reversals relative to the direction of aggregate income change:

$$S = \begin{cases} \{(Y^K, Y^P) : Y^K < Y^P\}, & \text{if} \quad \overline{Y}^K > \overline{Y}^P \\ \{(Y^K, Y^P) : Y^K > Y^P\}, & \text{if} \quad \overline{Y}^K < \overline{Y}^P \end{cases}. \tag{B.17}$$

It follows that this measure of exchange mobility is decreasing in the supermodular order and thus the concordance order with common marginals.

Cowell and Flachaire Measure The aggregate mobility measure is

$$\int\int \psi_\alpha\left(Y^K, Y^P\right) f\left(Y^K, Y^P\right) dY^K dY^P, \tag{B.18}$$

where the function $\psi_\alpha(x, y) = {}^1\!/\!_{\alpha(\alpha-1)}\,({}^x\!/\!_{\overline{x}})^\alpha\,({}^y\!/\!_{\overline{y}})^{1-\alpha}$ has the following cross-partial derivative:

$$\frac{\psi_\alpha(x, y)}{\partial x \partial y} = -\frac{1}{\overline{x}^\alpha \overline{y}^{1-\alpha}} x^{\alpha-1} y^{-\alpha} \leq 0\,. \tag{B.19}$$

Thus the function is submodular. The result then follows directly as concordance is equivalent to the supermodular order when marginals are constant (Tchen 1980).

Shorrocks Measure Note that the elements stochastic matrix $\mathcal{Q}$ are given by

$$\mathcal{Q}_{i,j} = Pr\left(\frac{j-1}{q} < R^K \leq \frac{j}{q} \mid \frac{i-1}{q} < R^P \leq \frac{i}{q}\right) \tag{B.20}$$

$$= \frac{1}{q}\left(C\left(\frac{i}{q}, \frac{j}{q}\right) - C\left(\frac{i-1}{q}, \frac{j}{q}\right) - C\left(\frac{i}{q}, \frac{j-1}{q}\right) + C\left(\frac{i-1}{q}, \frac{j-1}{q}\right)\right), \tag{B.21}$$

and the trace is

$$trace\,(\mathcal{Q}) = \sum_{i=1}^{q} \mathcal{Q}_{i,i}. \tag{B.22}$$

It follows that differences in the Shorrocks measure satisfy

$$\begin{aligned}
M^B - M^A \propto & trace\left(\mathcal{Q}^A\right) - trace\left(\mathcal{Q}^B\right)\\
&= \frac{1}{q}\sum_{i=1}^{q}\left(C^A\left(\tfrac{i}{q}, \tfrac{i}{q}\right) - C^A\left(\tfrac{i-1}{q}, \tfrac{i}{q}\right) - C^A\left(\tfrac{i}{q}, \tfrac{i-1}{q}\right) + C^A\left(\tfrac{i-1}{q}, \tfrac{i-1}{q}\right)\right)\\
&\quad - \frac{1}{q}\sum_{i=1}^{q}\left(C^B\left(\tfrac{i}{q}, \tfrac{i}{q}\right) - C^B\left(\tfrac{i-1}{q}, \tfrac{i}{q}\right) - C^B\left(\tfrac{i}{q}, \tfrac{i-1}{q}\right) + C^B\left(\tfrac{i-1}{q}, \tfrac{i-1}{q}\right)\right)\\
&= \frac{1}{q}\sum_{i=1}^{q}\left(\Delta C\left(\tfrac{i}{q}, \tfrac{i}{q}\right) - \Delta C\left(\tfrac{i-1}{q}, \tfrac{i}{q}\right) - \Delta C\left(\tfrac{i}{q}, \tfrac{i-1}{q}\right) + \Delta C\left(\tfrac{i-1}{q}, \tfrac{i-1}{q}\right)\right) \geq 0,
\end{aligned}$$

where the final inequality follows directly from Assumption 1. □

Proof of Proposition 5. This is immediate as an application of the closure properties of the concordance order. See Theorem 9.A.1 Shaked and Shanthikumar (2007). □

Proof of Corollary 1. The result follows immediately from the closure properties of the concordance order (See Theorem 9.A.1 Shaked and Shanthikumar 2007) □

Appendix C. Proofs for Section 4

Proof of Proposition 6. Since $C^A \succeq C^\dagger$ and M is assumed monotone in the concordance order we have $M(C^A) \leq M(C^\dagger)$. By uniform continuity of M we have

$$|M(C^\dagger) - M(C^B)| \leq \omega_M(||C^\dagger - C^B||_\infty) \leq \omega_M(\epsilon). \tag{C.1}$$

Therefore $M(C^A) \leq M(C^\dagger) \leq M(C^B) + \omega_M(\epsilon)$ which proves the first result. If M is Lipschitz continuous with constant L_M, then $\omega_M(\epsilon) \leq L_M \epsilon$ which proves the second result. □

Appendix D. Proofs for Section 6

Proof of Lemma 2. As $I^{*A}(Y^P) \geq I^{*B}(Y^P)$ for all Y^P (Assumption C) and f is increasing in I, we have

$$H^{K,A} \mid Y^P = y \quad \succeq_{\text{FOSD}} \quad H^{K,B} \mid Y^P = y \qquad \forall y.$$

Applying a common CDF (e.g. B), the rank variables satisfy

$$F_{R^{K,A}|Y^P}(r \mid y) \;\leq\; F_{R^{K,B}|Y^P}(r \mid y) \qquad \forall r \in (0,1),\ \forall y, \tag{D.1}$$

where r is the r^{th} quantile of $H^{K,B}$'s distribution.

Now consider the three permissable cases for $W(H^K)$ outline in the main-text:

Case 1: Rank-dependent wages ($F^A_W(w) = F^B_W(w)\ \forall w$). This implies the marginal distributions of wages are held fixed in the two economies. Integrating inequality (D.1) gives the bivariate joint density as Y^P has the same marginal density f_{Y^P} in both economies. Hence

$$(R^K, Y^P)^A \quad \succeq \quad (R^K, Y^P)^B.$$

Which is equivalent to the concordance ordering as the copula is preserved under monotone transformations.

Case 2: Common monotone-increasing function of human capital ($W^A(H^K) = W^B(H^K) = W(H^K)$). In this case, an immediate consequence of (D) is that child incomes in economy A first order stochastic dominate those in economy B.

Thus, by decomposing the supermodular order into marginals and copulas (Thm 2. Meyer and Strulovici 2013) we have an increase in concordance.

Case 3: Increased returns to human capital ($W^A(H^K) \geq W^B(H^K) \; \forall H^K$). Here analogous reasoning applies.

□

Proof of Corollary 4. For any Y^P, let $V(I;\lambda) = U\left(C^P\right) + \delta \mathbb{E}_\theta\left[U\left(W_\lambda(H^K)\right)\right]$ denote the value of the objective function for choice I and parameter λ controls the counterfactual. The proof follows from Topkis theorem under increasing differences. If the cross-partial (or discrete analogue) satisfies

$$\frac{\partial^2 V}{\partial I \partial \lambda} = \frac{\partial}{\partial \lambda}\left[\delta \frac{\partial f(I, H^P)}{\partial I} \mathbb{E}_\theta\left[U'\left(W_\lambda(H^K)\right) W'_\lambda(H^K)\right]\right] \geq 0, \tag{D.2}$$

then investment shifts outwards as the policy parameter λ increases. Under mild regularity conditions on general equilibrium effects, [31] it is easy to verify that the stated counterfactuals satisfy increasing differences as the production function is supermodular and the wage schedule is strictly increasing. Thus $I^*(Y^P;\lambda') \leq I^*(Y^P;\lambda) \quad$ whenever $\lambda' < \lambda$.

□

Appendix E. Construction of the Replication Dataset

The construction of the data follows Jácome, Kuziemko, and Naidu (2025). I briefly summarize the key features of their approach here and refer the interested reader to Jácome, Kuziemko, and Naidu (2025).

[31] If the wage schedule is constant across regimes, the partial derivative contributes zero. More generally, $\partial^2 V/\partial I \partial \lambda \geq 0$ holds under standard increasing-differences conditions—for instance, multiplicative wage shifts $W_\lambda(H) = g(\lambda) W_0(H)$ with $g'(\lambda) > 0$, or any change that makes $\mathbb{E}_\theta[U'(W_\lambda(H_K)) W'_\lambda(H_K)]$ increasing in λ. Policy counterfactuals in quantitative models (e.g., Abbott et al. 2019; Lee and Seshadri 2019) typically satisfy such conditions. See Topkis (1998) or Milgrom and Shannon (1994) for general treatments of increasing differences.

Data sources. I pool the fifteen U.S. surveys identified by Jácome, Kuziemko, and Naidu (2025) that report (i) respondents' current family income, (ii) father's occupation when the respondent was growing up, (iii) respondent race, and (iv) birthplace or childhood region (South vs. non-South). The surveys include American National Election Studies, General Social Survey, the Panel Study of Income Dynamics (specifically 1997 and 2017 waves), Occupational Changes in a Generation 1962 and 1973 surveys, the National Longitudinal Surveys, and others listed in Appendix E of their paper.

I use IPUMS census extracts from 1910 to 2019 (American Community Survey). These extracts differ from the original analysis which uses the full 1940 census available on the NBER server. They may also differ in other years due to differences in the size of the census-extracts selected.

Sample. I retain U.S-born men and women aged 30–50, the window that best approximates permanent income while minimising life-cycle bias . Respondents must have non-missing family income, race, region, and father's occupation. Foreign-born individuals are excluded because parental incomes are imputed from U.S. sources.

Respondent income. Each survey's family-income question is harmonised into 10–12 real-1950-dollar bins; continuous responses are recoded to the bin mid-point. Bottom-coded observations (including 0) are set to 0.75 × the upper boundary of this lowest bin and those in the top-bin are set to 1.25 × lower-bin boundary.

Parental income imputation. Fathers' (and, where available, mothers') occupations are mapped into 28 broad categories (e.g., skilled crafts, farm operators) . Predicted parental family income is the mean household income in matching occupation × race × South cells drawn from historical micro-data: 1901 Cost of Living Survey & 1900 Census (early cohorts), full-count 1940 Census plus the 1936 Expenditure Survey, and 1960–1990 Censuses . For farmers and self-employed, incomes are adjusted following Collins and Wanamaker (2022); non-working fathers receive values imputed from contemporaneous Census means. Father race is proxied by respondent race; father region by respondent childhood region.

Weighting. Where surveys supply sampling weights, I re-centre them to mean 1. I then adjust all surveys so that each birth-decade cell matches Census race-sex population shares (white men/women, Black men/women); surveys lacking weights receive weight equal to 1 before adjustment.

Cohorts. Decade birth cohorts 1910s–1970s are used. For each respondent (a) log family income and (b) percentile rank in the pooled income distribution are computed; parental income is treated analogously, yielding comparable distributions of income ranks across cohorts.

These steps reproduce the long-run, nationally representative parent–child dataset used in the original study and permit the replication of Figure 1 in their paper (Panel (A), Figure 8 here).